Building and evaluation of a PBPK model for Ketoconazole in adults

Version	**v1.0-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Ketoconazole-Model/releases/tag/v1.0-OSP12.2
OSP Version	12.2
Qualification Framework Version	3.3

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

Table of Contents

- 1 Introduction
- 2 Methods
  - 2.1 Modeling strategy
  - 2.2 Data used
  - 2.3 Model parameters and assumptions
- 3 Results and Discussion
  - 3.1 Ketoconazole final input parameters
  - 3.2 Diagnostic plots
    - 3.2.1 Ketoconazole goodness-of-fit diagnostics
    - 3.2.2 N-deacetylketoconazole goodness-of-fit diagnostics
  - 3.3 Concentration-time profiles
    - 3.3.1 Model Building
    - 3.3.2 Model Verification
- 4 Conclusion
- 5 References

1 Introduction

Ketoconazole is an imidazole antifungal and a strong inhibitor of CYP3A4 and P-glycoprotein (P-gp, ABCB1). Although oral ketoconazole use is restricted because of safety concerns, ketoconazole remains a clinically important reference perpetrator for CYP3A4-mediated and transporter-mediated drug-drug interaction assessments.

The model describes ketoconazole together with the circulating metabolites N-deacetylketoconazole and N-deacetyl-N-hydroxyketoconazole. The parent-metabolite structure was published by Marok 2023 to describe oral ketoconazole pharmacokinetics and quantify the contribution of parent and metabolites to CYP3A4 and P-gp inhibition. This report evaluates only the ketoconazole and metabolite pharmacokinetic data included in the model publication.

2 Methods

2.1 Modeling strategy

The general strategy for model development was based on a middle-out PBPK workflow. Drug-dependent parameters were taken from experimental data or estimated using established in silico methods where possible. Parameters that could not be fixed from independent evidence were optimized against clinical concentration-time profiles and then evaluated using independent oral dosing scenarios.

The model was implemented as a parent-metabolite model for ketoconazole, N-deacetylketoconazole and N-deacetyl-N-hydroxyketoconazole. Ketoconazole plasma profiles from oral solution, capsule, and tablet studies were used to inform absorption, dissolution and disposition parameters, while the Weiss et al. clinical data provided the only plasma concentration-time profiles for N-deacetylketoconazole Weiss 2022.

Ketoconazole absorption was evaluated across fasted and fed dosing conditions, including solution, capsule, and tablet formulations. The model includes CYP3A4-mediated metabolism, arylacetamide deacetylase-mediated N-deacetylketoconazole formation, UGT1A4-mediated metabolism, FMO3-mediated metabolism of both metabolites, P-gp transport, and competitive inhibition of CYP3A4 and P-gp by ketoconazole and both metabolites. The resulting model structure supports both concentration-time profile evaluation and DFI or DDI perpetrator simulations Marok 2023.

2.2 Data used

In vitro and physicochemical data

Table 1 gives the drug-dependent parameters used in the final model. The table notes document corrections to values in Table 1 and Supplement Table S1.3 of Marok 2023. Optimized values are identified as such. Assumed values are traced to the basis stated in the supplement.

Parameter	Unit	Value	Source	Description
Ketoconazole				
MW	g/mol	531.43	Chemicalize 2021	Molecular weight.
logP	-	2.52	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	1	Heel 1982	Fraction unbound in plasma.
pK_a,base	-	2.94	Chemicalize 2021	pKa of a basic ionization site.
pK_a,base	-	6.51	Chemicalize 2021	pKa of a basic ionization site.
Solubility	mg/L	2.033×10^4 (pH 1.2); 4.3×10^2 (pH 3); 1.0×10^2 (pH 5); 7.00 (pH 6.8); 6.00 (pH 7.5); 5.40 (deionized water)[a]	Ghazal 2015, Table 1	Aqueous solubility in the specified media.
Density	g/cm^3	1.40	ChemSpider 2021	Solid-state density.
D_{aq}	dm^2/min	3.75×10^{-7}	Optimized	Aqueous diffusion coefficient.
P_{int}, fasted	cm/min	1.56×10^{-5}	Optimized	Specific transcellular intestinal permeability in the fasted state.
P_{int}, fed	cm/min	9.95×10^{-6}	Optimized	Specific transcellular intestinal permeability in the fed state.
GET, fasted	min	15	OSP Suite Manual 2018	Gastric emptying time in the fasted state.
GET, fed	min	45	Fisher 1987	Gastric emptying time in the fed state.
Cellular permeabilities	-	PK-Sim Standard	OSP Suite Manual 2018	Cellular permeabilities calculated with the PK-Sim Standard method.

Parameter	Unit	Value	Source	Description
Partition coefficients	-	Berezhkovskiy	Berezhkovskiy 2004	Tissue-to-plasma partition coefficients calculated with the Berezhkovskiy method.
GFR fraction	-	1	Assumed	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1	Assumed	Fraction released continuously into bile.
$K_{m,AADAC}$	µmol/L	1.88	Fukami 2016	Michaelis constant for AADAC-mediated hydrolysis.
$k_{cat,AADAC}$	min^{-1}	0.87	Optimized	Catalytic rate constant for AADAC-mediated hydrolysis.
$K_{m,CYP3A4}$	µmol/L	0.00846	Weiss 2022	Michaelis constant for CYP3A4 metabolism, assumed equal to $K_{i,CYP3A4}$.
$k_{cat,CYP3A4}$	min^{-1}	0.10	Optimized	Catalytic rate constant for CYP3A4-mediated metabolism.
$K_{m,UGT1A4}$	µmol/L	7	Bourcier 2010	Michaelis constant for UGT1A4-mediated glucuronidation.
$k_{cat,UGT1A4}$	min^{-1}	0.31	Optimized	Catalytic rate constant for UGT1A4-mediated glucuronidation.
$K_{m,P-gp}$	µmol/L	0.035	Weiss 2022	Michaelis constant for P-glycoprotein transport, assumed equal to $K_{i,P-gp}$.
$k_{cat,P-gp}$	min^{-1}	0.33	Optimized	Catalytic rate constant for P-glycoprotein transport.

Parameter	Unit	Value	Source	Description
$K_{i,CYP3A4}$	µmol/L	0.00846	Weiss 2022	Inhibition constant for CYP3A4, derived from IC_{50}.
$K_{i,P\text{-}gp}$	µmol/L	0.035	Weiss 2022	Inhibition constant for P-gp.
Particle radius	µm	11.749; 111.06; 205.46[b]	Elder 2007	Particle radii assigned to the specified dose fractions.
N-deacetylketoconazole				
MW	g/mol	489.40	Chemicalize 2021b	Molecular weight.
logP	-	3.75	Optimized	Octanol-water partition coefficient of the neutral species.
f_u	%	1	Heel 1982	Fraction unbound in plasma.
pK_a,base	-	0.20	Chemicalize 2021b	pKa of a basic ionization site.
pK_a,base	-	6.42	Chemicalize 2021b	pKa of a basic ionization site.
pK_a,base	-	8.90	Chemicalize 2021b	pKa of a basic ionization site.
Solubility	mg/L	1.24×10^3	Chemicalize 2021b	Aqueous solubility at pH 6.5.
Cellular permeabilities	-	Charge-dependent Schmitt	Kawai 1994	Cellular permeabilities calculated with the charge-dependent Schmitt method.
Partition coefficients	-	Rodgers and Rowland	Rodgers 2006	Tissue-to-plasma partition coefficients calculated with the Rodgers and Rowland method.

Parameter	Unit	Value	Source	Description
GFR fraction	-	1	Assumed	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1	Assumed	Fraction released continuously into bile.
$K_{m,FMO3}$	µmol/L	1.17[d]	Rodriguez 1997	Michaelis constant for FMO3-mediated N-oxidation.
$k_{cat,FMO3}$	min^{-1}	378.65	Optimized	Catalytic rate constant for FMO3-mediated N-oxidation.
$K_{i,CYP3A4}$	µmol/L	0.022	Weiss 2022	Inhibition constant for CYP3A4, derived from IC_{50}.
$K_{i,P-gp}$	µmol/L	0.119	Weiss 2022	Inhibition constant for P-gp.
N-deacetyl-N-hydroxyketoconazole				
MW	g/mol	505.40	Chemicalize 2022	Molecular weight.
logP	-	4.20	Chemicalize 2022	Octanol-water partition coefficient of the neutral species.
f_u	%	1	Heel 1982	Fraction unbound in plasma.
pK_a,base	-	3.42	Chemicalize 2022	pKa of a basic ionization site.
pK_a,base	-	6.42	Chemicalize 2022	pKa of a basic ionization site.
Solubility	mg/L	4.40×10^{3}[c]	Marok 2023, Table 1	Aqueous solubility at pH 6.5.
P_{organ}	cm/min	0	Assumed	Specific organ permeability.

Parameter	Unit	Value	Source	Description
Cellular permeabilities	-	Charge-dependent Schmitt	Kawai 1994	Cellular permeabilities calculated with the charge-dependent Schmitt method.
Partition coefficients	-	Berezhkovskiy	Berezhkovskiy 2004	Tissue-to-plasma partition coefficients calculated with the Berezhkovskiy method.
GFR fraction	-	1	Assumed	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1	Assumed	Fraction released continuously into bile.
CL_{FMO3}	L/(µmol·min)	0.09	Optimized	FMO3-mediated clearance.
$K_{i,CYP3A4}$	µmol/L	0.022	Weiss 2022	Inhibition constant for CYP3A4, assumed equal to the N-deacetylketoconazole value.
$K_{i,P\text{-}gp}$	µmol/L	0.119[e]	Weiss 2022	Inhibition constant for P-gp.

Table 1: Drug-dependent parameters used in the final ketoconazole model. AADAC: arylacetamide deacetylase. FMO3: flavin-containing monooxygenase 3. GFR: glomerular filtration rate. P-gp: P-glycoprotein. UGT1A4: uridine diphosphate glucuronosyltransferase 1A4. [a] Parent ketoconazole values were converted from mg/mL in Table 1 of Ghazal 2015 to mg/L. Table 1 of Marok 2023 gives an incorrect pH 3 value and assigns pH 7 to the deionized-water value. [b] Supplement Table S1.3 incorrectly labels the particle radii as nm. The model values are equivalent to the listed µm values. [c] The value is 4.40×10^3 mg/L, equivalent to 4.40 mg/mL, as reported in Table 1 of Marok 2023. Supplement Table S1.3 is incorrect. [d] Table 1 of Marok 2023 states 1.77 µmol/L. Supplement Table S1.3 states 1170 nmol/L, which is equal to the model value. [e] Assumed equal to the N-deacetylketoconazole value.

Clinical data

Clinical plasma concentration-time profiles were taken from the published studies compiled for the model publication (Marok 2023). The evaluation includes 53 oral profiles across solution, tablet, capsule, fasted, fed, single-dose, and multiple-dose conditions (Table 2). Seven profiles were used for

model building, and 46 profiles were used for model verification. This assignment differs from the training and test assignment in the model publication. When a source publication also describes an interaction study, this compound report includes only ketoconazole or metabolite pharmacokinetics and does not evaluate victim-drug exposure. The model file contains five more observed-data sets that this report does not evaluate. Three are plasma profiles without a simulation in the model file. Two are the fractions excreted to faeces (Daneshmend 1988) and urine (Heel 1982).

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Boyce 2012	200, multiple dose, profile 1, fasted	26.6 (18–39)	73.5 (53.8–98.8)	41.67% female	24	Tablet	EM
Boyce 2012	200, multiple dose, profile 2, fasted	26.6 (18–39)	73.5 (53.8–98.8)	41.67% female	24	Tablet	EM
Chin 1995	200, fasted	22–41	NR	33.34% female	9	Tablet	EM
Craven 1983	1200, multiple dose, fed	NR	NR	NR	2	Tablet	EM
Craven 1983	800, multiple dose, fed	NR	NR	NR	2	Tablet	EM
Daneshmend 1981	200, fed	NR	NR	NR	6	Tablet	EM
Daneshmend 1981	400, fed	NR	NR	NR	NR	Tablet	EM
Daneshmend 1983	200, fasted	25 (21–46)	NR	Male	8	Tablet	EM
Daneshmend 1983	200, multiple dose, fasted	25 (21–46)	NR	Male	8	Tablet	EM
Daneshmend 1984	200, fed	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM
Daneshmend 1984	200, profile 1, fasted	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM
Daneshmend 1984	400, fasted	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM
Daneshmend 1984	400, fed	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM
Daneshmend 1984	600, fasted	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM
Daneshmend 1984	600, fed	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM
Daneshmend 1984	800, fasted	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Daneshmend 1984	800, fed	23 (20–31)	64 (50–75)	62.5% female	8	Tablet	EM
Greenblatt 1998	200 twice daily for 5 days, fed	18–38	NR	NR	8	Tablet	EM
Heel 1982	100, fasted	NR	NR	NR	12	Tablet	EM
Heel 1982	200, fasted	NR	NR	NR	12	Tablet	EM
Heel 1982+	200, fasted	NR	NR	NR	12	Solution	EM
Heel 1982	400, fasted	NR	NR	NR	12	Tablet	EM
Huang 1986	200, fasted	20 (18–25)	76.4 (61.2–95.3)	Male	23	Tablet	EM
Huang 1986+	200, profile 1, fasted	20 (18–25)	76.4 (61.2–95.3)	Male	12	Solution	EM
Huang 1986+	200, profile 2, fasted	20 (18–25)	76.4 (61.2–95.3)	Male	23	Solution	EM
Huang 1986+	400, fasted	20 (18–25)	76.4 (61.2–95.3)	Male	12	Solution	EM
Huang 1986+	800, fasted	20 (18–25)	76.4 (61.2–95.3)	Male	12	Solution	EM
Knupp 1993	200, fasted	30 (24–36)	78.8	Male	12	Tablet	EM
Männistö 1982	200, fed	24 (22–26)	62 (55–70)	50% female	10	Tablet	EM
Männistö 1982	200, profile 1, fasted	24 (22–26)	62 (55–70)	50% female	10	Tablet	EM
Männistö 1982	200, profile 2, fasted	24 (22–26)	62 (55–70)	50% female	10	Tablet	EM
Männistö 1982	200, profile 3, fasted	24 (22–26)	62 (55–70)	50% female	10	Tablet	EM
Männistö 1982	200, with juice, fasted	24 (22–26)	62 (55–70)	50% female	10	Tablet	EM

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Patel 2011	200, multiple dose, fasted	36 (22–43)	74.7 (50.1–95)	19% female	15	Tablet	EM
Piscitelli 1991	400, fasted	18–30	NR	Male	6	Tablet	EM
Polk 1999	400, fasted	23 (19–41)	77.4 (64.2–99.8)	Male	12	Tablet	EM
Sadeghnia 2005	400, profile 1, fasted	23–29	59–78	Male	12	Tablet	EM
Sadeghnia 2005	400, profile 2, fasted	23–29	59–78	Male	12	Tablet	EM
Sekar 2008	200 twice daily for 4 days, fed	NR	NR	NR	NR	Tablet	EM
Solomon 2007	400, profile 1, fasted	23.2 (18–45)	NR	Male	24	Tablet	EM
Solomon 2007	400, profile 2, fasted	23.2 (18–45)	NR	Male	24	Tablet	EM
Sriwiriyajan 2007+	400, fasted	33.7 (22–55)	NR	75% female	12	Capsule	EM
Tiseo 1998+	200 once daily for 7 days, fasted	NR	NR	NR	21	Tablet	EM
U.S. FDA 1998	200, profile 1, fasted	NR	NR	NR	23	Tablet	EM
U.S. FDA 1998	200, profile 1, fed	NR	NR	NR	23	Tablet	EM
U.S. FDA 1998	200, profile 2, fasted	NR	NR	NR	39	Tablet	EM
U.S. FDA 1998	200, profile 2, fed	NR	NR	NR	23	Tablet	EM
U.S. FDA 1998	200, profile 3, fasted	NR	NR	NR	39	Tablet	EM
Van der Meer 1980	200, control, fasted	28–42	NR	NR	3	Tablet	EM

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Weiss 2022	400, fasted	27.34 (20–48)	74.44 (57.5–100)	Male	12	Tablet	EM
Wire 2007	200, multiple dose, fasted	NR	NR	NR	15	Tablet	EM
Yuen 1999	200, profile 1, fasted	NR	NR	NR	18	Tablet	EM
Yuen 1999	200, profile 2, fasted	NR	NR	NR	18	Tablet	EM

Table 2: Clinical ketoconazole concentration-time profiles used for model building and verification. *: Single oral dose unless otherwise specified; EM: extensive metabolizer; NR: not reported; $^{+}$: data used for model building. EM is the model default when study-specific CYP2D6 information is not available. Food conditions are retained in the dose and schedule field. Craven 1983 studied patients with fungal infections of the central nervous system. Knupp 1993 studied HIV-seropositive patients. Sriwiriyajan 2007 studied HIV-infected patients. The Sekar 2008 simulation starts twice-daily dosing at 48 h and gives 9 doses up to the morning dose on day 7.

2.3 Model parameters and assumptions

Absorption and formulations

The model includes oral solution and tablet applications. The capsule study is simulated with the tablet particle-size distribution. Solution scenarios were used to separate systemic disposition from dissolution-limited tablet absorption. Tablet absorption was described with formulation-specific dissolution and particle-size assumptions, with fed-state scenarios used to describe the reduced and delayed absorption observed for weak-base ketoconazole under altered gastrointestinal conditions Marok 2023, Table 1.

Fed-state settings are a gastric emptying time of 45 min and the fed intestinal permeability. The model publication applied them to studies with reported food intake. As an assumption, it also applied them when the food state was not reported and the observed T_{max} was more than 2 h, when more than one dose was given per day, or when a single tablet dose was 800 mg or higher Marok 2023. Four profiles that Table 2 labels as fasted use the fed gastric emptying time: Boyce 2012 (9) and (10), Daneshmend 1984 800 mg (23), and Wire 2007 (94). Three of these profiles also use the fed intestinal permeability. Boyce 2012 (10) uses the fasted intestinal permeability. The twice-daily profiles Daneshmend 1983 (29) and Patel 2011 (65) use fasted settings.

Distribution

Ketoconazole is a weak base with high plasma protein binding and pH-dependent solubility. Distribution parameters were based on physicochemical properties and PBPK tissue partitioning methods, with optimized lipophilicity where required to reproduce systemic exposure. The metabolites

were represented as separate compounds with their own physicochemical properties and protein-binding assumptions Marok 2023, Table 1.

Metabolism, transport and inhibition

Ketoconazole metabolism includes formation of N-deacetylketoconazole by arylacetamide deacetylase and further metabolism through CYP3A4 and UGT pathways. Ketoconazole and metabolites were represented as CYP3A4 and P-gp (ABCB1) inhibitors. In the model publication, this structure described DDI perpetrator behavior more accurately than a parent-only inhibition model Marok 2023, Table 1, and Weiss 2022.

3 Results and Discussion

The parent-metabolite PBPK model for ketoconazole was developed and evaluated with clinical pharmacokinetic data after oral administration. The evaluation covers doses from 100 to 1200 mg, single-dose and multiple-dose regimens, solution and solid oral formulations, fasted and fed conditions, and plasma observations for ketoconazole and N-deacetylketoconazole.

Model-building studies included solution data from Heel 1982 and Huang 1986, a multiple-dose tablet study from Tiseo 1998, and a 400 mg capsule study from Sriwiriyajan 2007. Verification used tablet studies across the evaluated dose, regimen, and food conditions. N-deacetylketoconazole observations were available from Weiss 2022.

The model includes formulation-dependent absorption, arylacetamide deacetylase-mediated N-deacetylketoconazole formation, CYP3A4 and UGT1A4 metabolism of ketoconazole, FMO3 metabolism of both metabolites, P-gp transport, and competitive CYP3A4 and P-gp inhibition by ketoconazole and its metabolites. Goodness-of-fit diagnostics are separated for ketoconazole and N-deacetylketoconazole because metabolite observations are limited to one study.

The next sections show:

1. the final model input parameters for the building blocks in Section 3.1.
2. analyte-specific goodness-of-fit diagnostics in Section 3.2.
3. simulated vs. observed concentration-time profiles for model building and verification in Section 3.3.

Interpretation is limited by the absence of intravenous data, estimated formulation and food-effect parameters, incomplete CYP3A4 and P-gp kinetic information, and sparse metabolite observations.

3.1 Ketoconazole final input parameters

The following tables are generated from the model and summarize the final compound input parameters for ketoconazole, N-deacetylketoconazole and N-deacetyl-N-hydroxyketoconazole. Interpret optimized parameters together with the sources in Section 2.2 and the assumptions in Section 2.3.

The particle-radius unit reported in Table S1.3 of the supplement is erroneous. The intended radii are 11.749, 111.06 and 205.46 µm for tablet dissolution bins 1, 2 and 3, respectively. The corresponding model values of 0.011749, 0.11106 and 0.20546 mm are correct. The solubility value shown below represents a pH-dependent table function rather than a single constant.[1]

Compound: Ketoconazole

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility table	12872590000000 mg/l	Publication-In Vitro-Extrapolated from observed data	Ghazal 2015	True
Lipophilicity	2.5243939623 Log Units	Parameter Identification-Parameter Identification-Optimized	fit	True
Fraction unbound (plasma, reference value)	0.01	Publication-Other	Heel 1982	True
Specific intestinal permeability (transcellular)	1.5597039897E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Fit fasted	True
Specific intestinal permeability (transcellular)	9.9497068191E-06 cm/min	Parameter Identification-Parameter Identification-Optimized	Fit fed	False
Cl	2	Internet-Other-Chemicalize		
Is small molecule	Yes			
Molecular weight	531.43 g/mol	Internet-Other-Chemicalize		
Plasma protein binding partner	Albumin			
Treat precipitated drug as	Insoluble	Unknown		
Aqueous diffusion coefficient	3.7482784124E-07 dm^2/min	Parameter Identification-Parameter Identification-Optimized		
Density (drug)	1.4 g/cm^3	Internet-Other-ChemSpider		
Enable supersaturation	Yes	Unknown		

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	PK-Sim Standard

Processes

Metabolizing Enzyme: CYP3A4-Assumed

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	0.008461538 µmol/l	Other-Assumption-Assumed from Ki value from Weiss 2022
kcat	0.1020204084 1/min	Parameter Identification-Parameter Identification-Optimized

Transport Protein: ABCB1-Assumed

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Transporter concentration	1 µmol/l	
Vmax	0 µmol/l/min	
Km	0.035 µmol/l	Other-Assumption-Assumed from Ki value from Weiss 2022
kcat	0.326999715 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-GFR

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: UGT1A4-Bourcier 2010

Molecule: UGT1A4

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	0 nmol/min/pmol rec. enzyme	
Km	7 µmol/l	Publication-In Vitro-recombinant enzymes
kcat	0.3062809592 1/min	Parameter Identification-Parameter Identification-Optimized

Inhibition: CYP3A4-Weiss 2022

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Ki	0.008461538 µmol/l	Publication-In Vitro-Calculated from IC50 and corrected for fumic

Metabolizing Enzyme: AADAC-Fukami 2016

Molecule: AADAC

Metabolite: N-deacetylketoconazole

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	28.3 pmol/min/mg mic. protein	
Km	1.8797 µmol/l	Publication-In Vitro-Corrected with $f_{u,mic}$ (17.5*0.107)
kcat	0.8694012741 1/min	Parameter Identification-Parameter Identification-Optimized

Inhibition: ABCB1-Weiss 2022

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Ki	0.035238095 µmol/l	Publication-In Vitro-Calculated from IC50 value

Compound: N-deacetylketoconazole

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1.24 mg/ml	Internet-Other-Chemicalize	Chemicalize	True
Reference pH	6.5	Internet-Other-Chemicalize	Chemicalize	True
Lipophilicity	3.7501327278 Log Units	Parameter Identification-Parameter Identification-Optimized	Fit	True
Fraction unbound (plasma, reference value)	1 %	Other-Assumption-assumed from value for Ketoconazole	Assumed	True
Cl	2	Internet-Other-Chemicalize		
Is small molecule	Yes			
Molecular weight	489.4 g/mol	Internet-Other-Chemicalize		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	Charge dependent Schmitt

Processes

Inhibition: CYP3A4-Weiss 2022

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Ki	0.022 µmol/l	Publication-In Vitro-Calculated from IC50 and corrected for fumic

Systemic Process: Glomerular Filtration-GFR

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: FMO3-Rodriguez 1997

Molecule: FMO3

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	1.167401215 µmol/l	Publication-In Vitro-recombinant enzymes; model value is correct, publication value is erroneous
kcat	378.6506200467 1/min	Parameter Identification-Parameter Identification-Optimized

Inhibition: ABCB1-Weiss 2022

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Ki	0.119047619 µmol/l	Publication-In Vitro-Calculated from IC50

Compound: N-deacetyl-N-hydroxyketoconazole

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	4.4 mg/ml	Internet-Other-Chemicalize	Chemicalize	True
Reference pH	6.5	Internet-Other-Chemicalize	Chemicalize	True
Lipophilicity	4.2 Log Units	Internet-Other-Chemicalize	Chemicalize	True
Fraction unbound (plasma, reference value)	1 %	Publication-Assumption-assumed from value for Ketoconazole	Assumed	True
Permeability	0 cm/min		Assumed	True
Cl	2	Internet-Other-Chemicalize		
Is small molecule	Yes			
Molecular weight	505.4 g/mol	Internet-Other-Chemicalize		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	Charge dependent Schmitt

Processes

Inhibition: CYP3A4-Weiss 2022

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Ki	0.022 µmol/l	Other-Assumption-assumed from value for N-deacetylketoconazole

Systemic Process: Glomerular Filtration-GFR

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Inhibition: ABCB1-Weiss 2022

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Ki	0.119047619 µmol/l	Other-Assumption-assumed from value for N-deacetylketoconazole

Metabolizing Enzyme: FMO3-Clearance

Molecule: FMO3

Parameters

Name	Value	Value Origin
In vitro CL/recombinant enzyme	0 µl/min/pmol rec. enzyme	
CLspec/[Enzyme]	0.0893329451 l/µmol/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

The goodness-of-fit diagnostics are separated by analyte. This prevents the parent and metabolite observations from sharing one legend and keeps the error measures analyte specific.

3.2.1 Ketoconazole goodness-of-fit diagnostics

Ketoconazole parent plasma observations are mapped to the ketoconazole plasma output.

Table 3-1: GMFE for Ketoconazole goodness-of-fit diagnostics

Group	GMFE
Ketoconazole	1.70

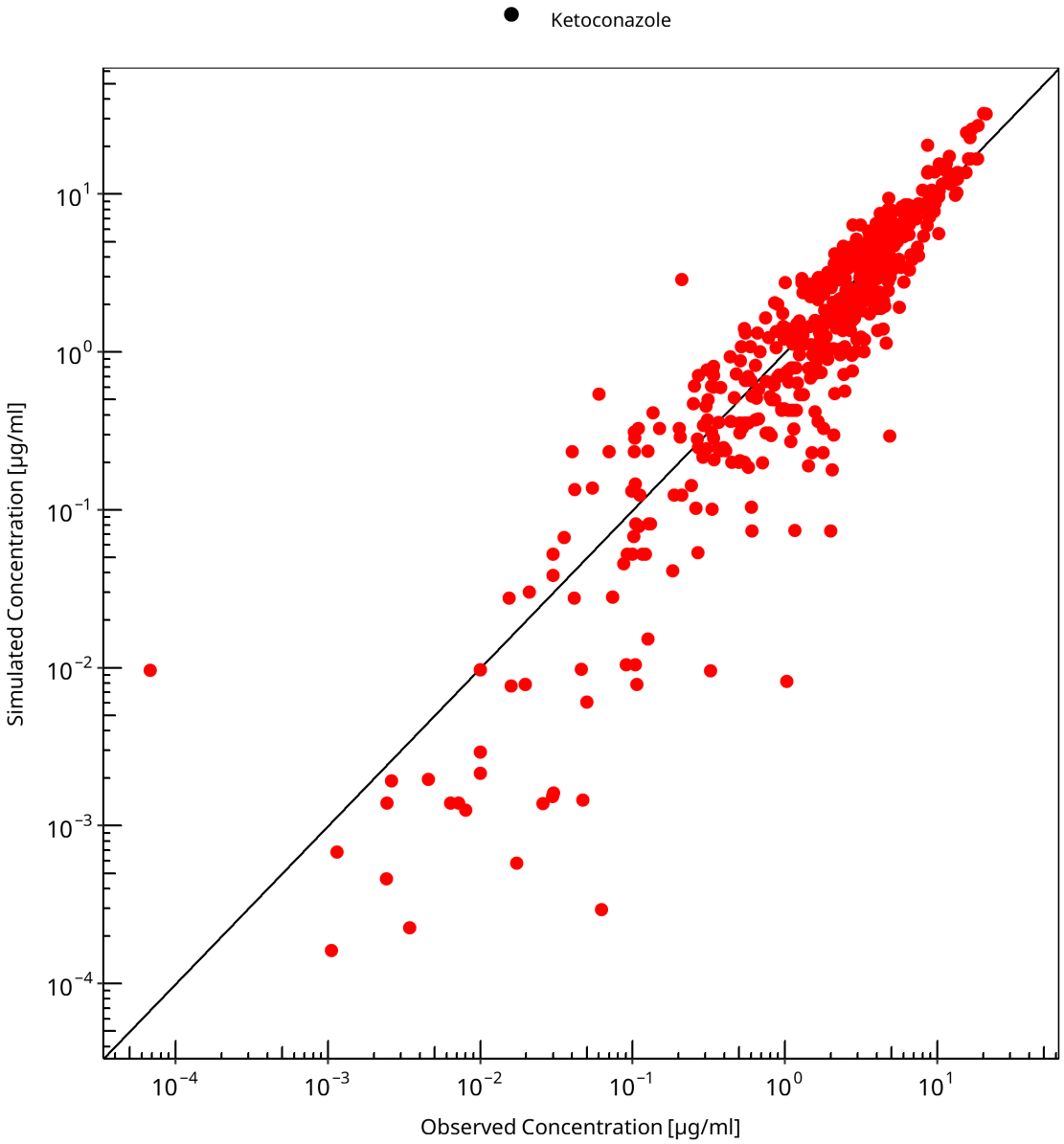


Figure 3-1: Ketoconazole goodness-of-fit diagnostics

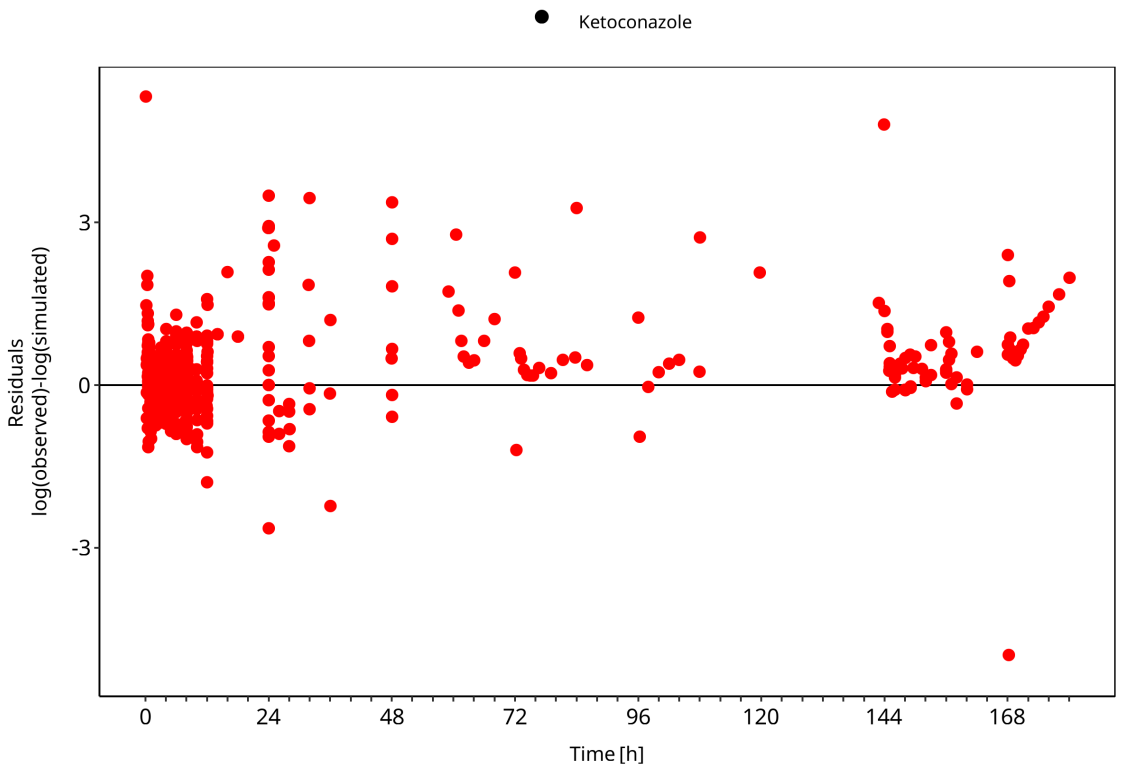


Figure 3-2: Ketoconazole goodness-of-fit diagnostics

3.2.2 N-deacetylketoconazole goodness-of-fit diagnostics

N-deacetylketoconazole plasma observations are mapped to the metabolite plasma output.

Table 3-2: GMFE for N-deacetylketoconazole goodness-of-fit diagnostics

Group	GMFE
N-deacetylketoconazole	1.93

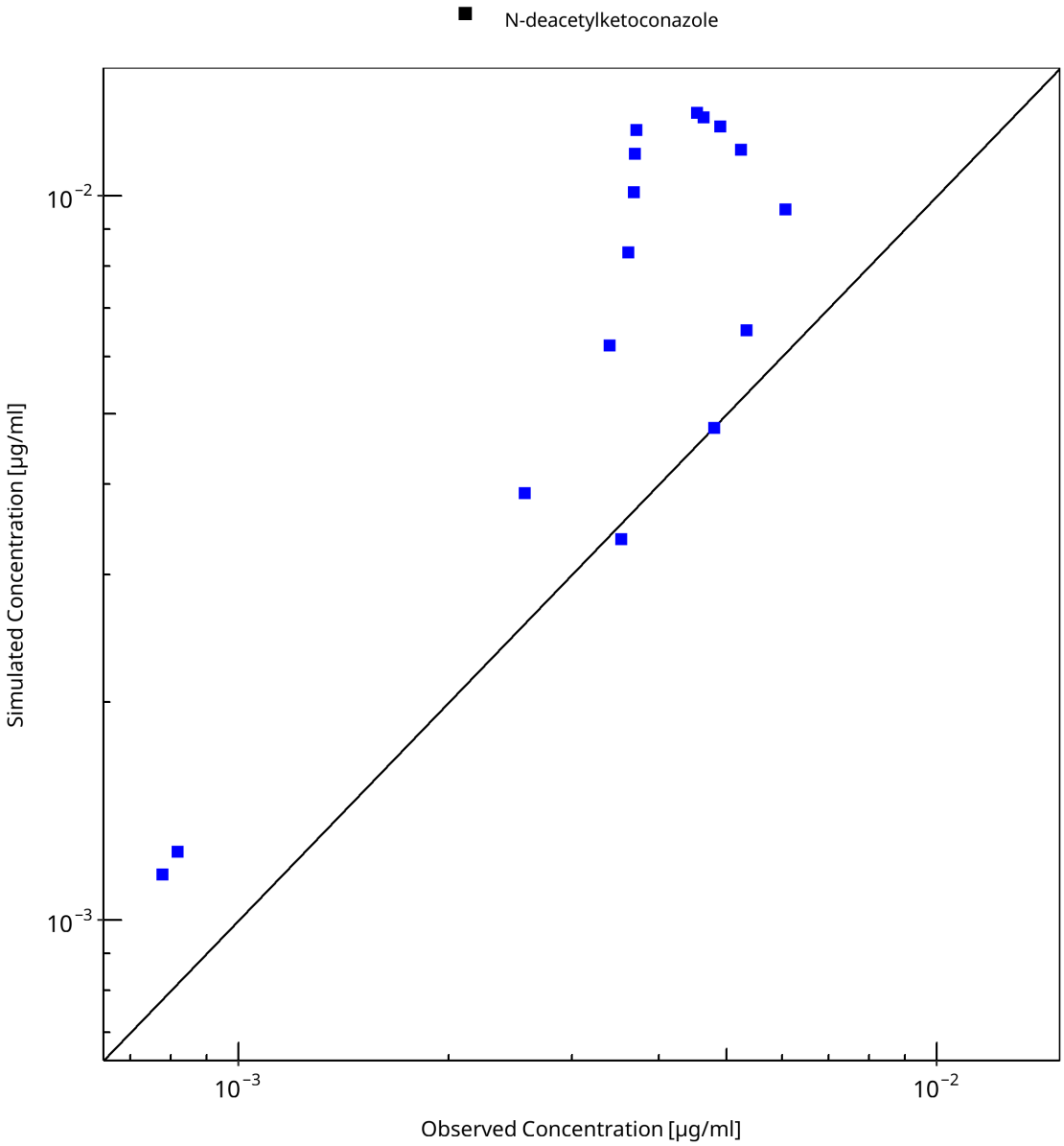


Figure 3-3: N-deacetylketoconazole goodness-of-fit diagnostics

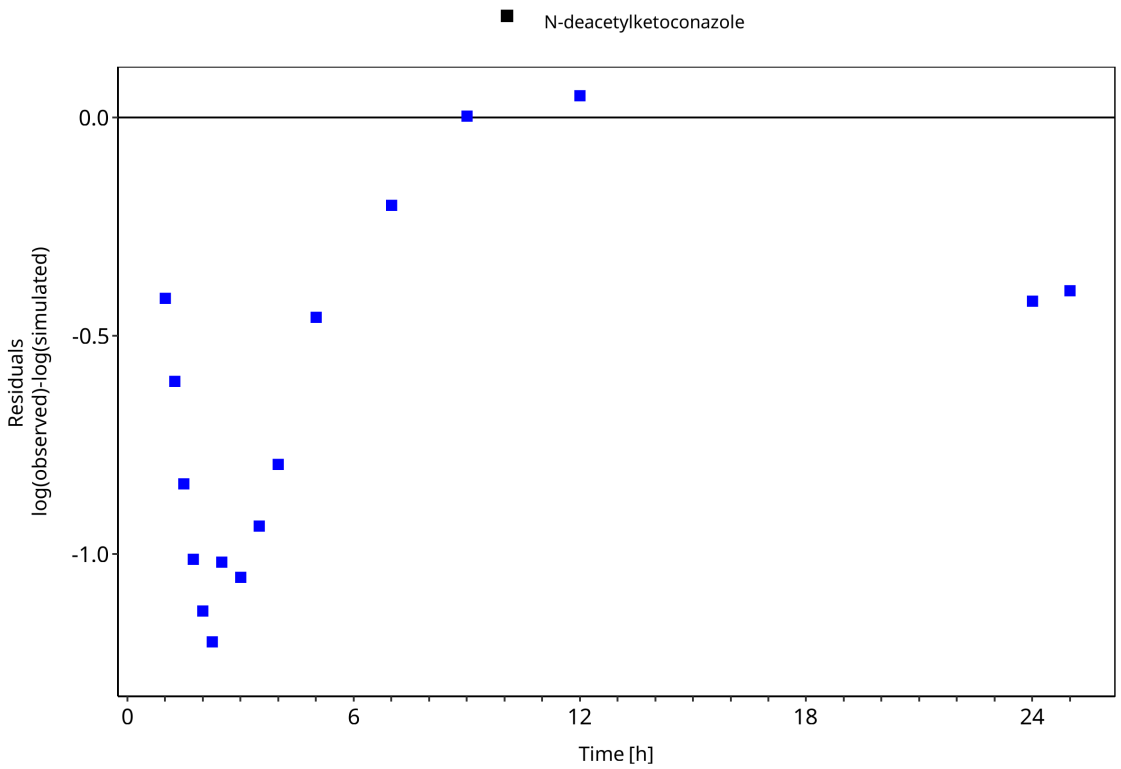


Figure 3-4: N-deacetylketoconazole goodness-of-fit diagnostics

3.3 Concentration-time profiles

Concentration-time profiles are grouped into model building and model verification. Table 2 gives the dose, schedule, food condition, and formulation of each profile.

3.3.1 Model Building

The model-building profiles cover solution and solid oral applications used to inform absorption, formulation, and disposition parameters.

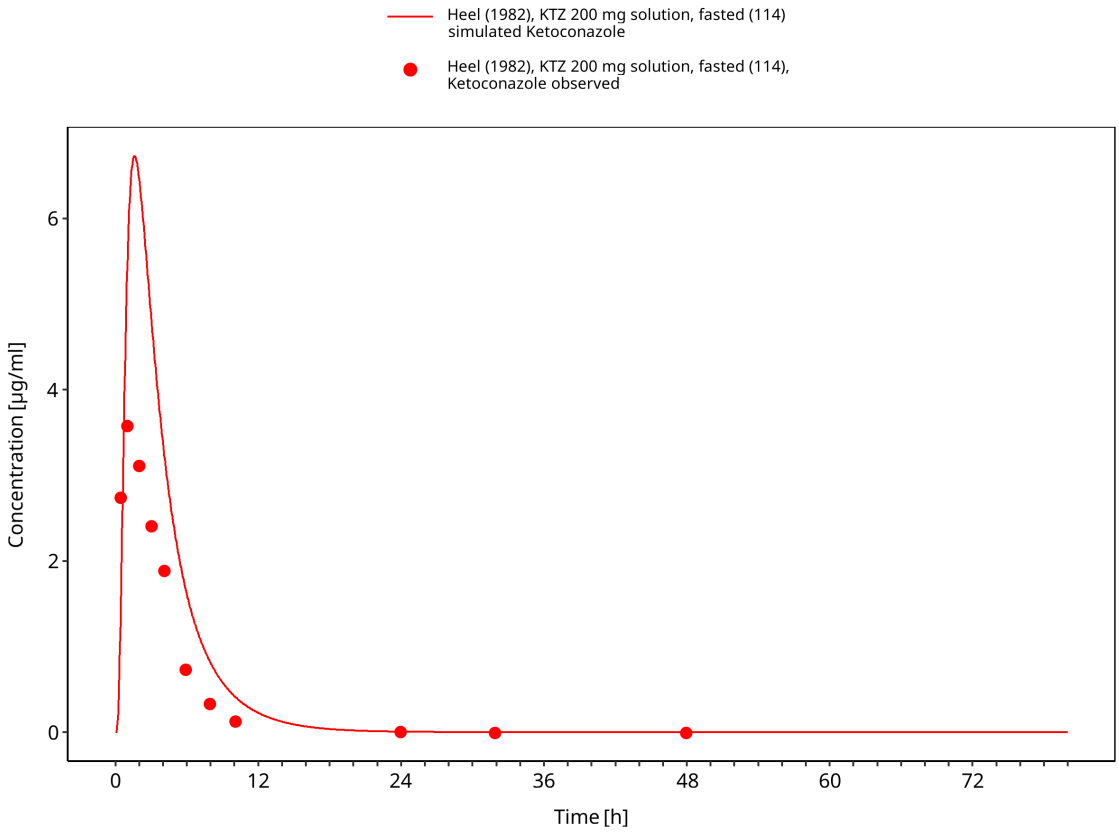


Figure 3-5: Heel 1982: KTZ 200 mg solution, fasted (114)

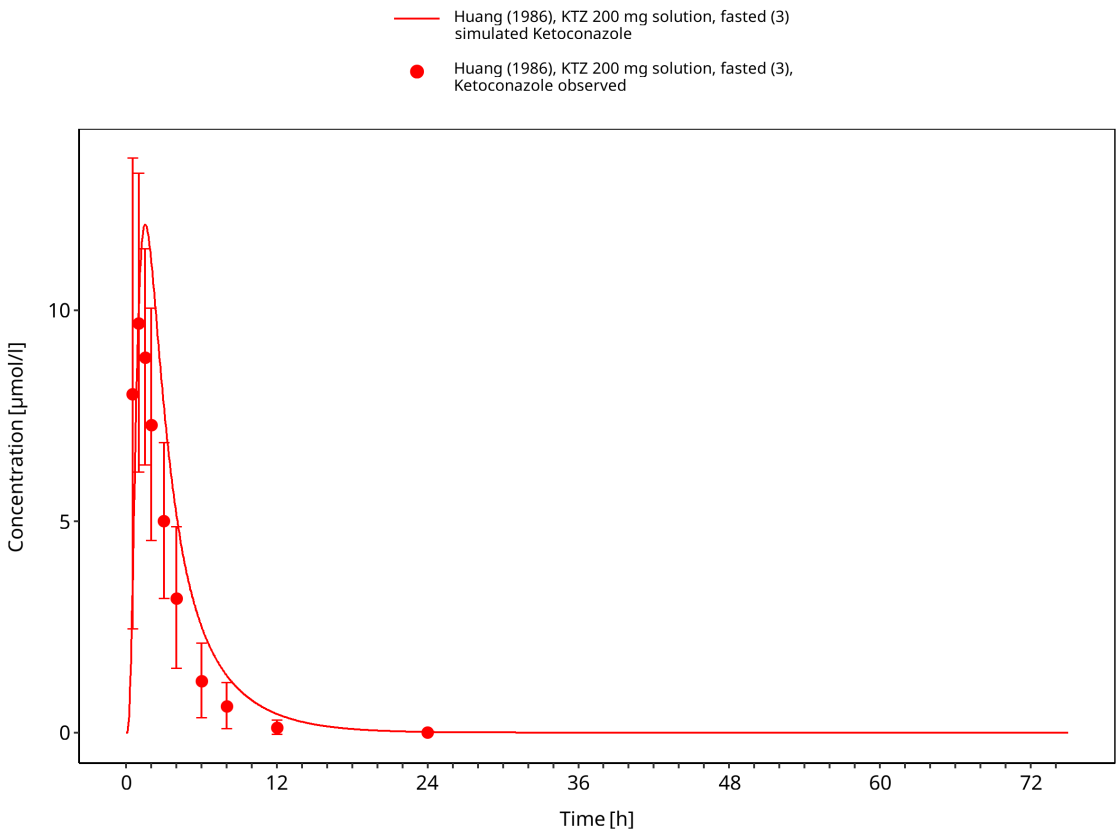


Figure 3-6: Huang 1986: KTZ 200 mg solution, fasted (3)

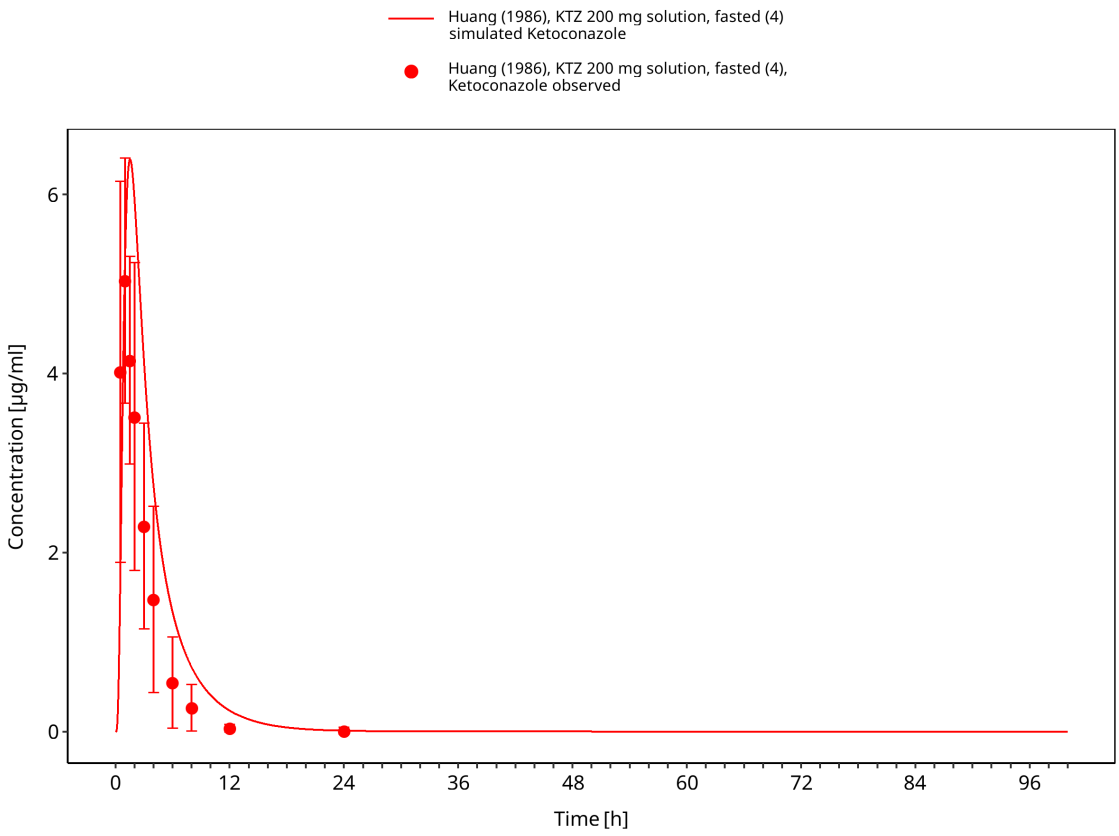


Figure 3-7: Huang 1986: KTZ 200 mg solution, fasted (4)

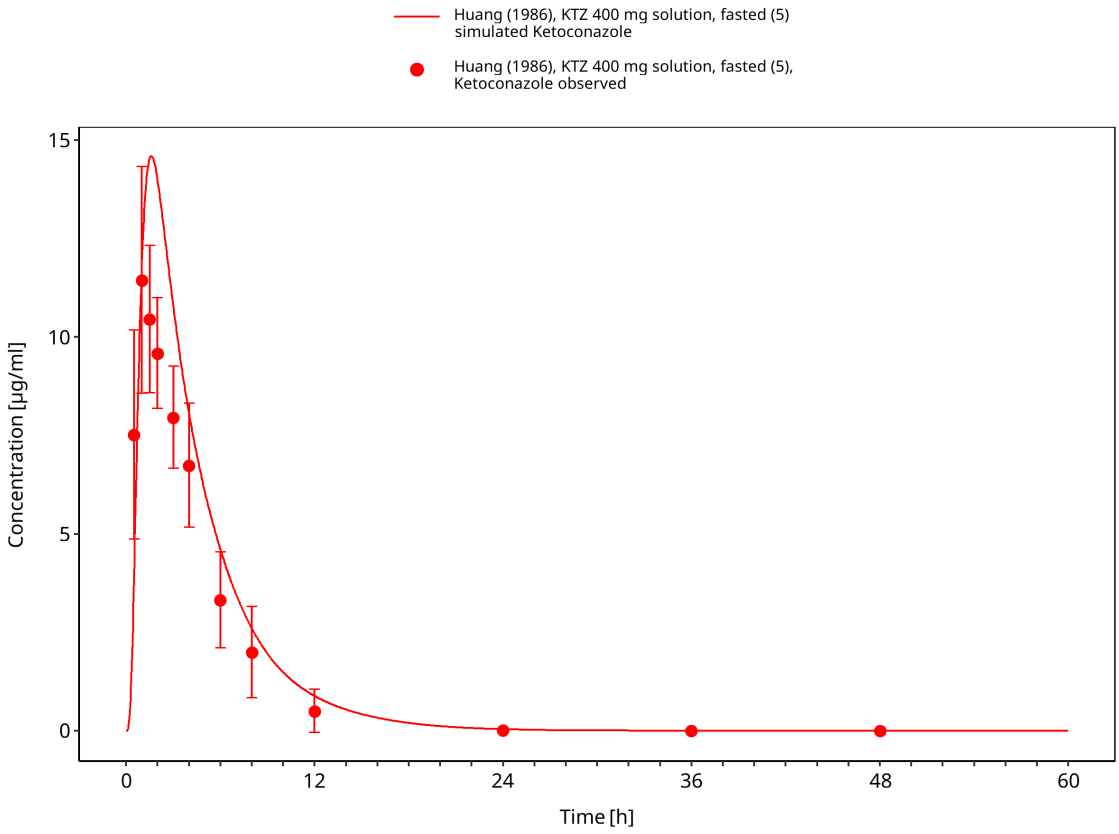


Figure 3-8: Huang 1986: KTZ 400 mg solution, fasted (5)

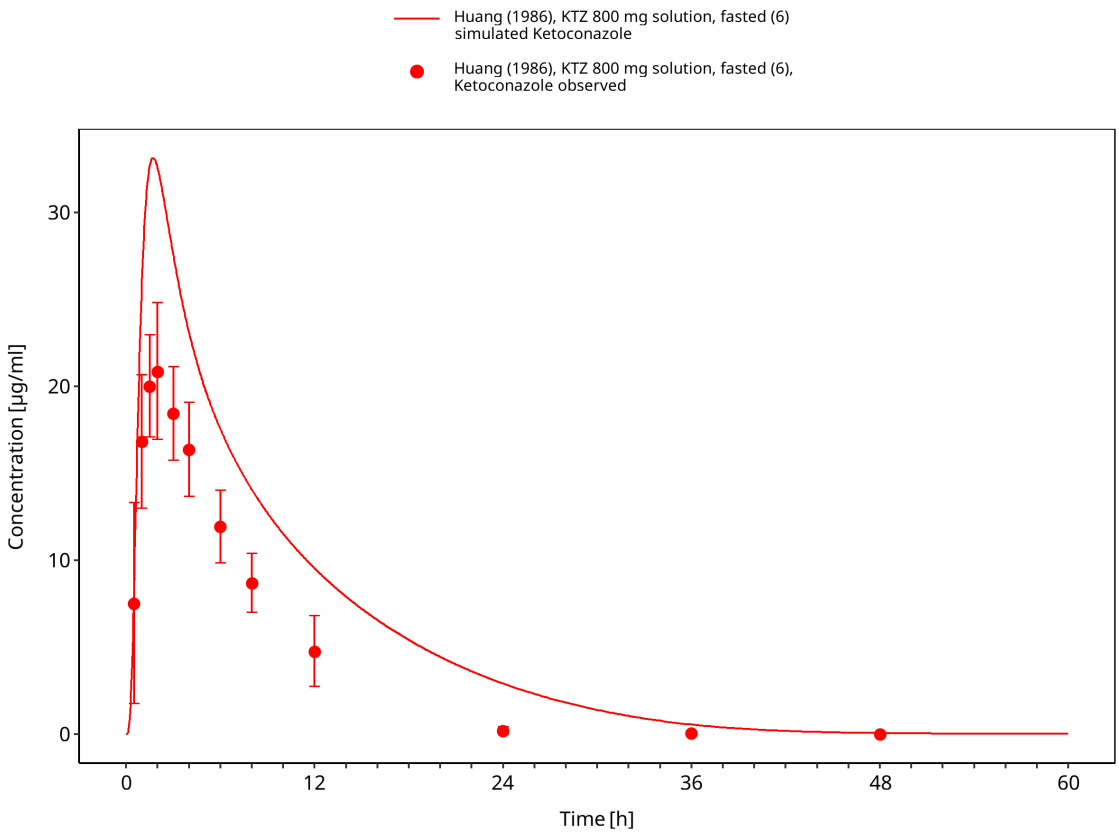


Figure 3-9: Huang 1986: KTZ 800 mg solution, fasted (6)

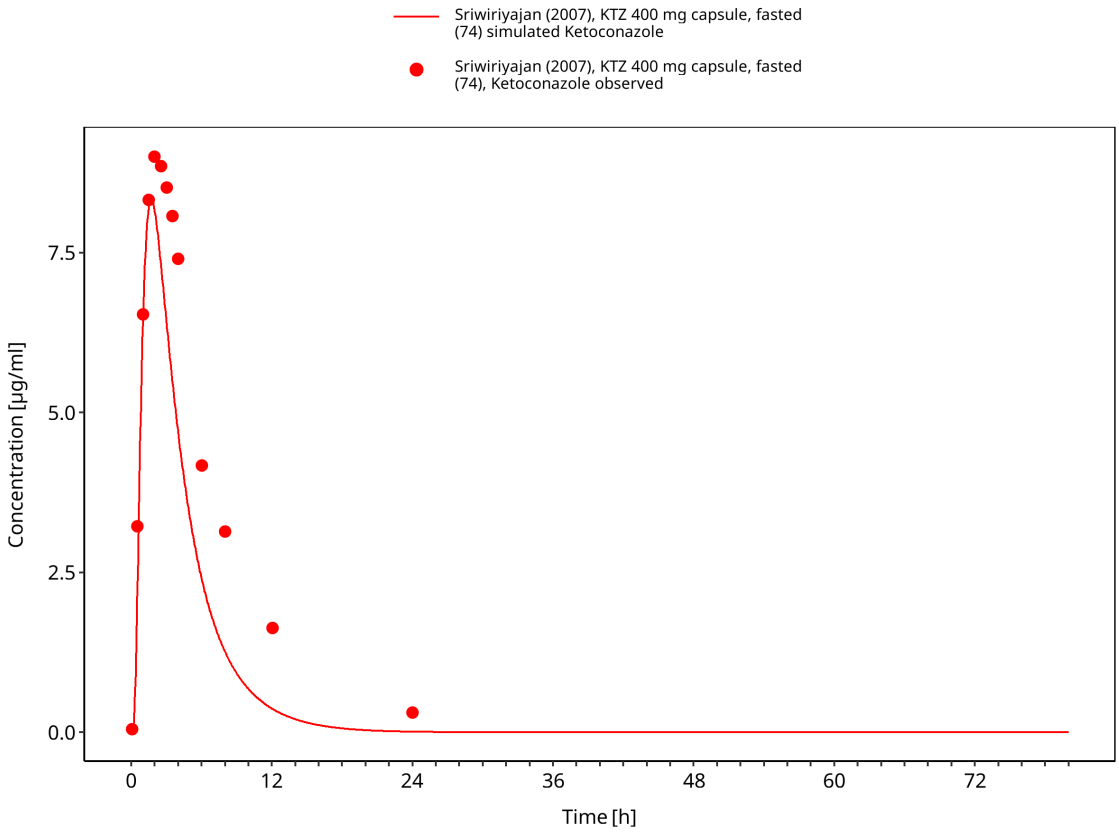


Figure 3-10: Sriwiriyajan 2007: KTZ 400 mg capsule, fasted (74)

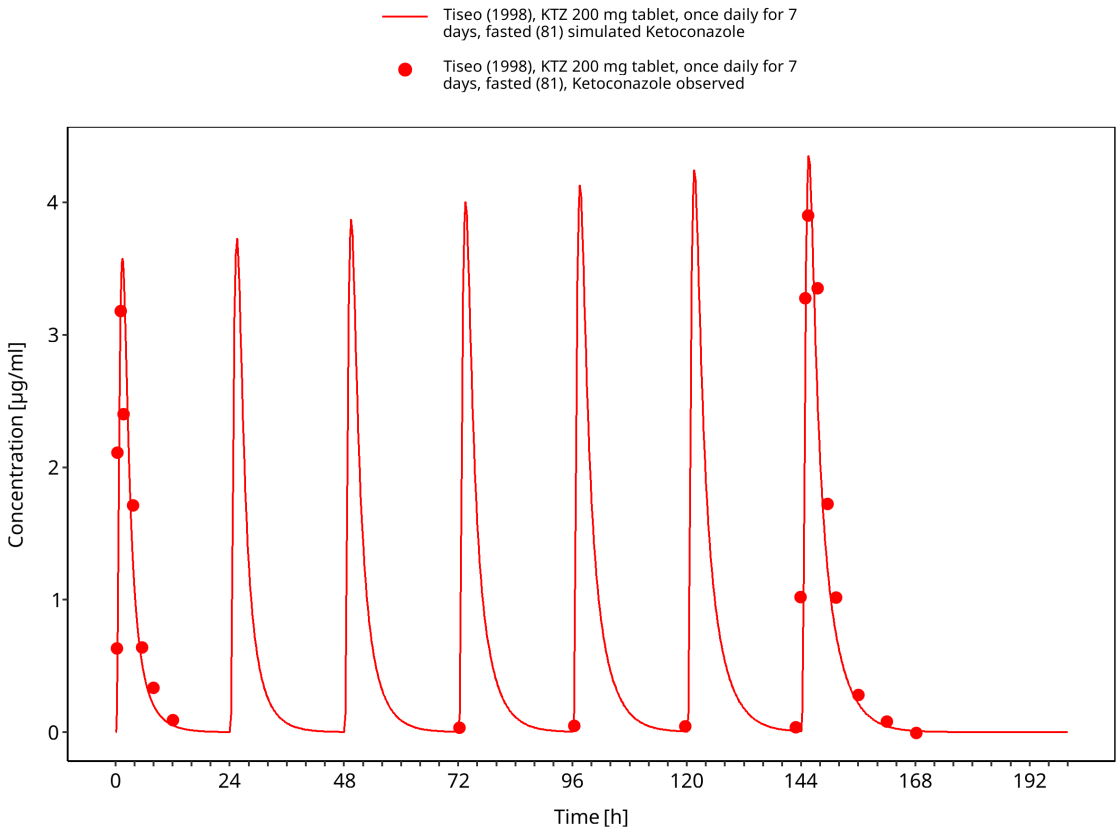


Figure 3-11: Tiseo 1998: KTZ 200 mg tablet q.d. 7d, fasted (81)

3.3.2 Model Verification

The model-verification profiles cover independent fasted, fed, single-dose, and multiple-dose oral studies.

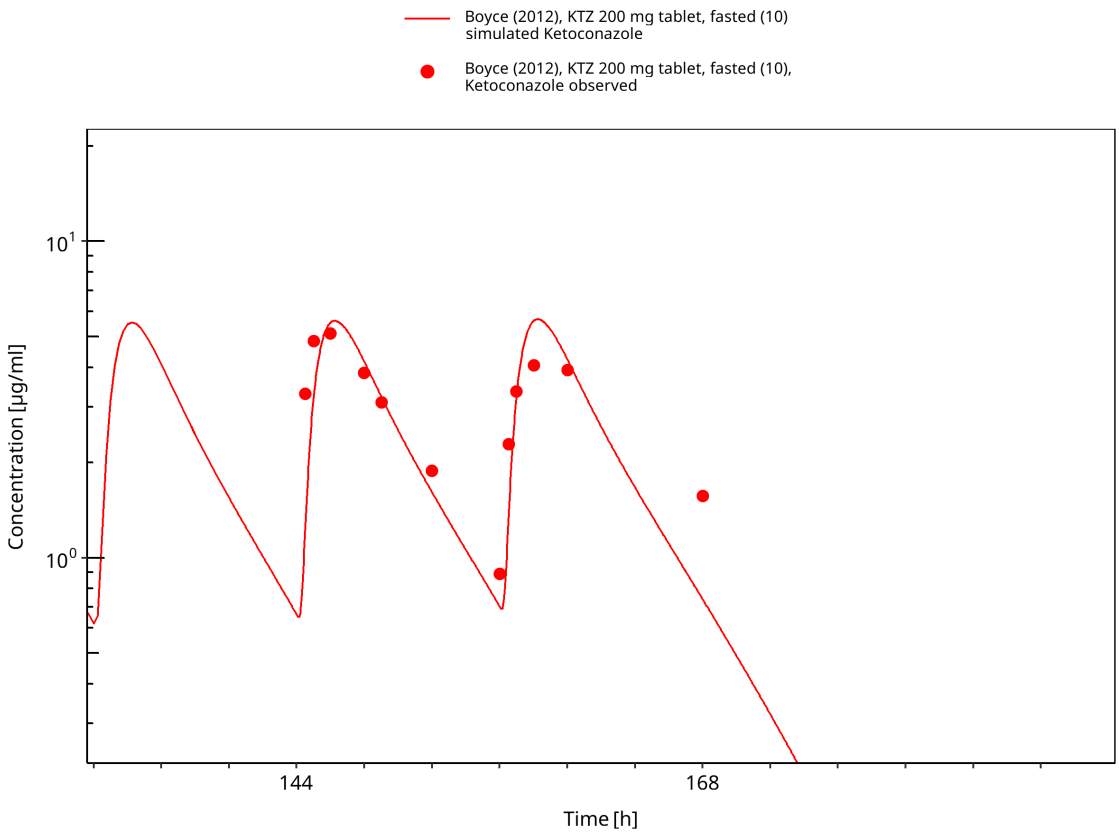


Figure 3-12: Boyce 2012: KTZ 200 mg tablet b.i.d. 7d, fasted (10)

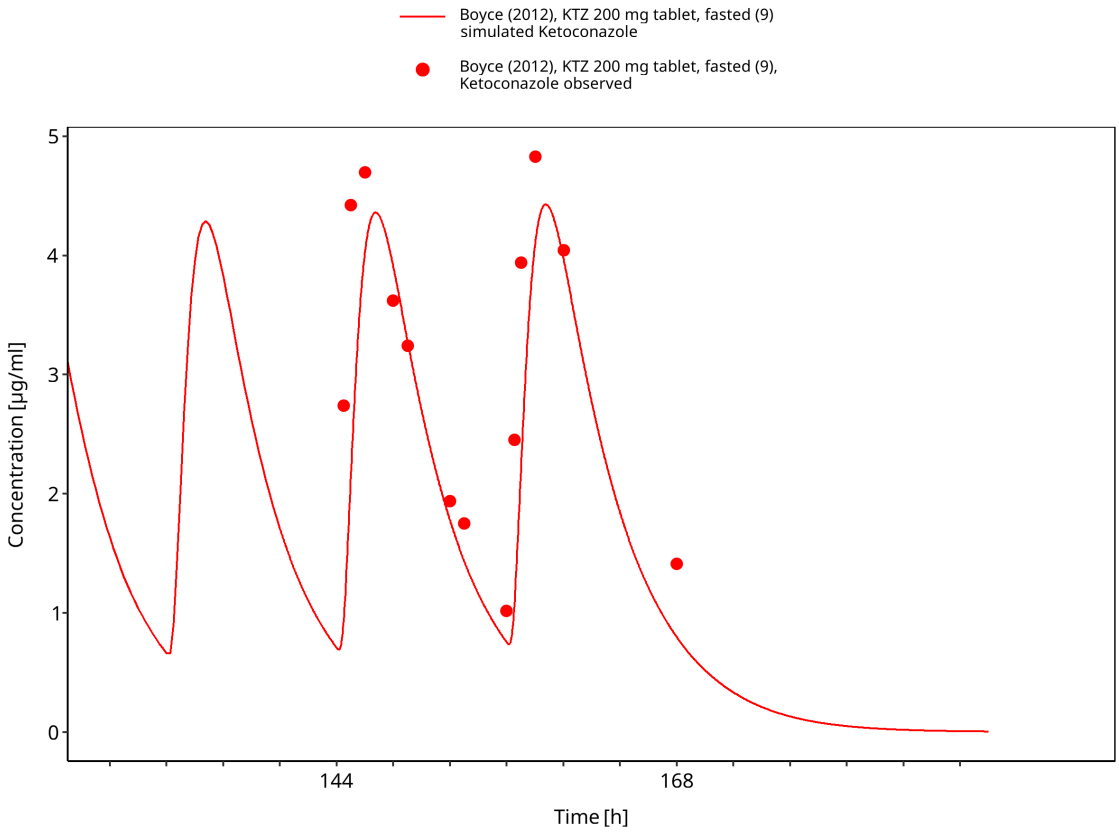


Figure 3-13: Boyce 2012: KTZ 200 mg tablet b.i.d. 7d, fasted (9)

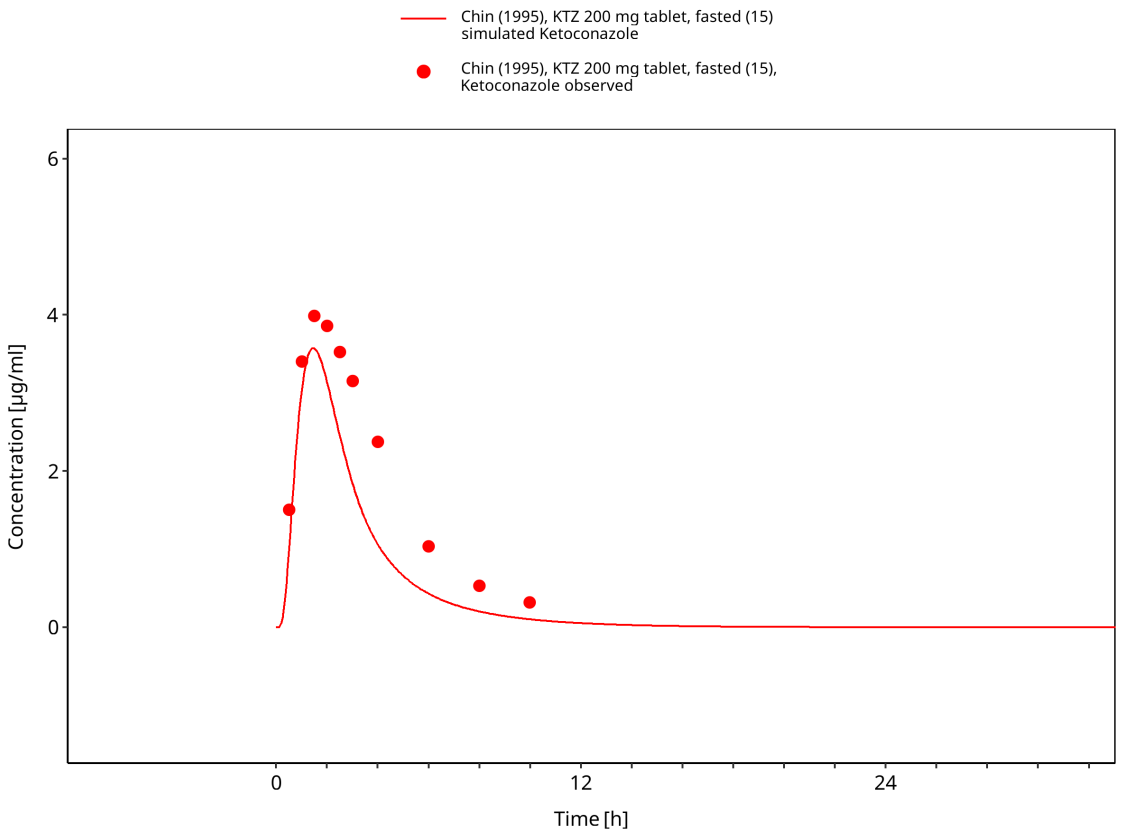


Figure 3-14: Chin 1995: KTZ 200 mg tablet, fasted (15)

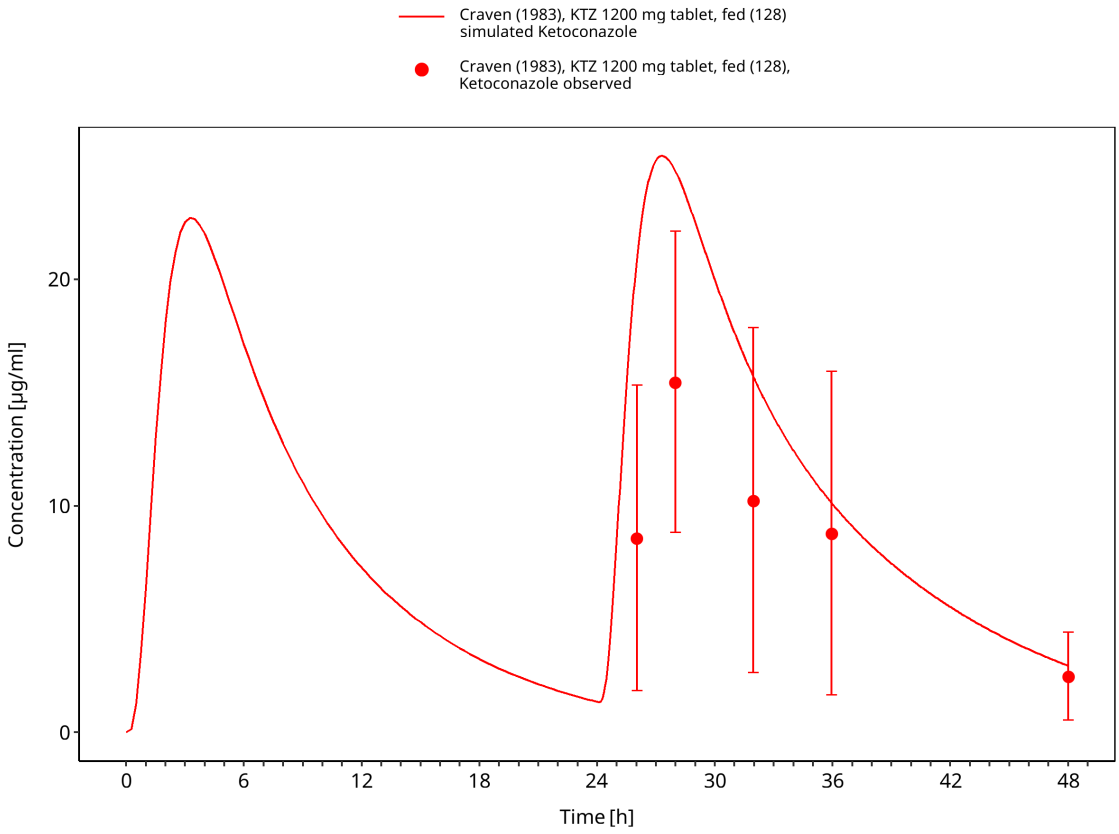


Figure 3-15: Craven 1983: KTZ 1200 mg tablet q.d. 2d, fed (128)

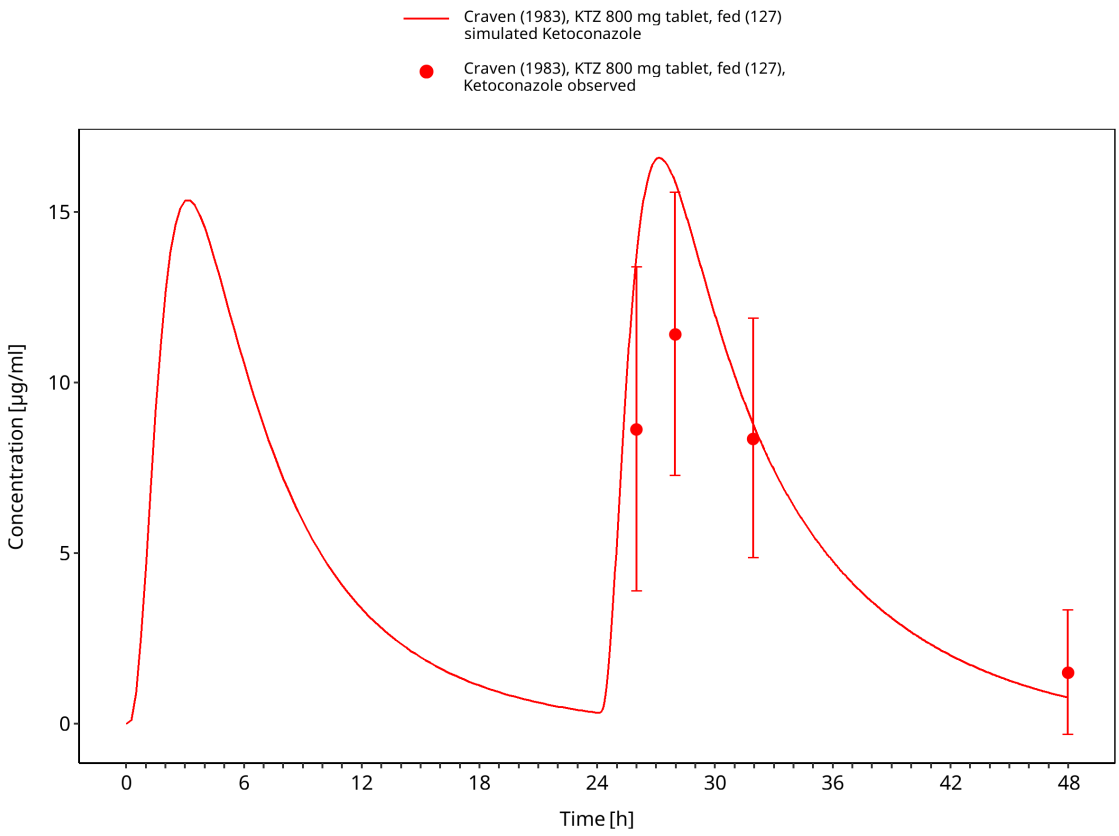


Figure 3-16: Craven 1983: KTZ 800 mg tablet q.d. 2d, fed (127)

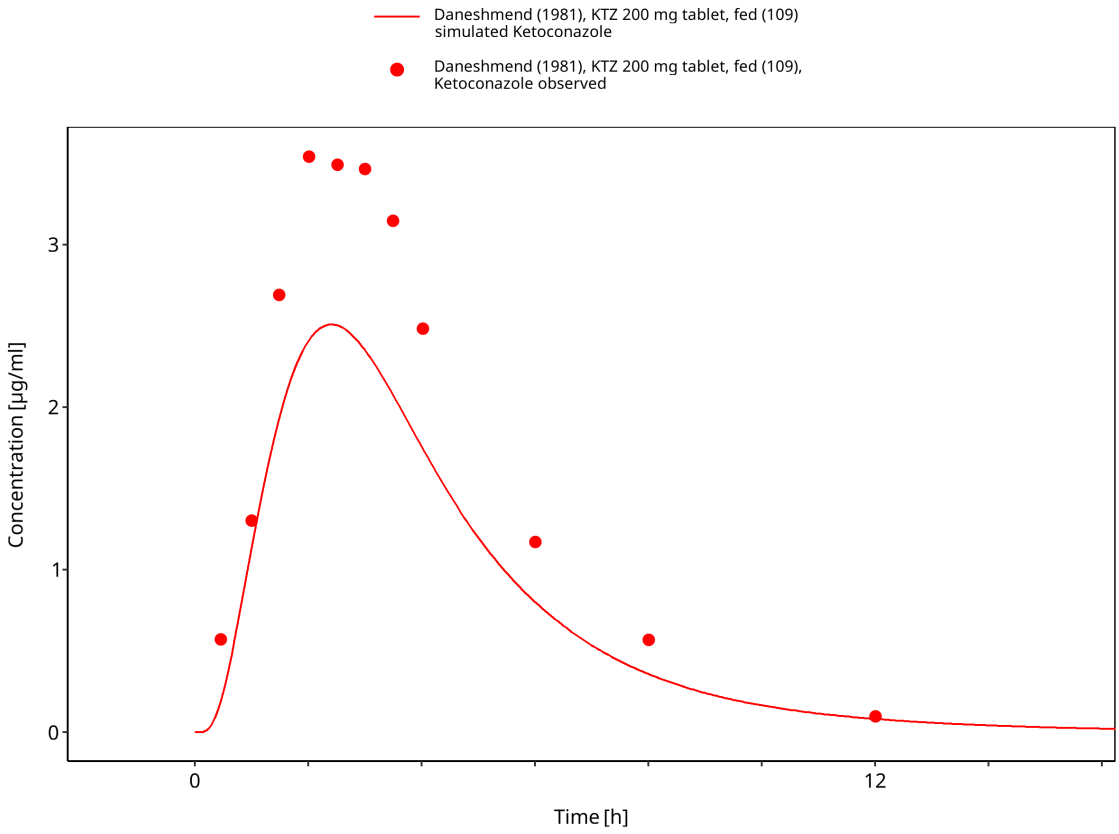


Figure 3-17: Daneshmend 1981: KTZ 200 mg tablet, fed (109)

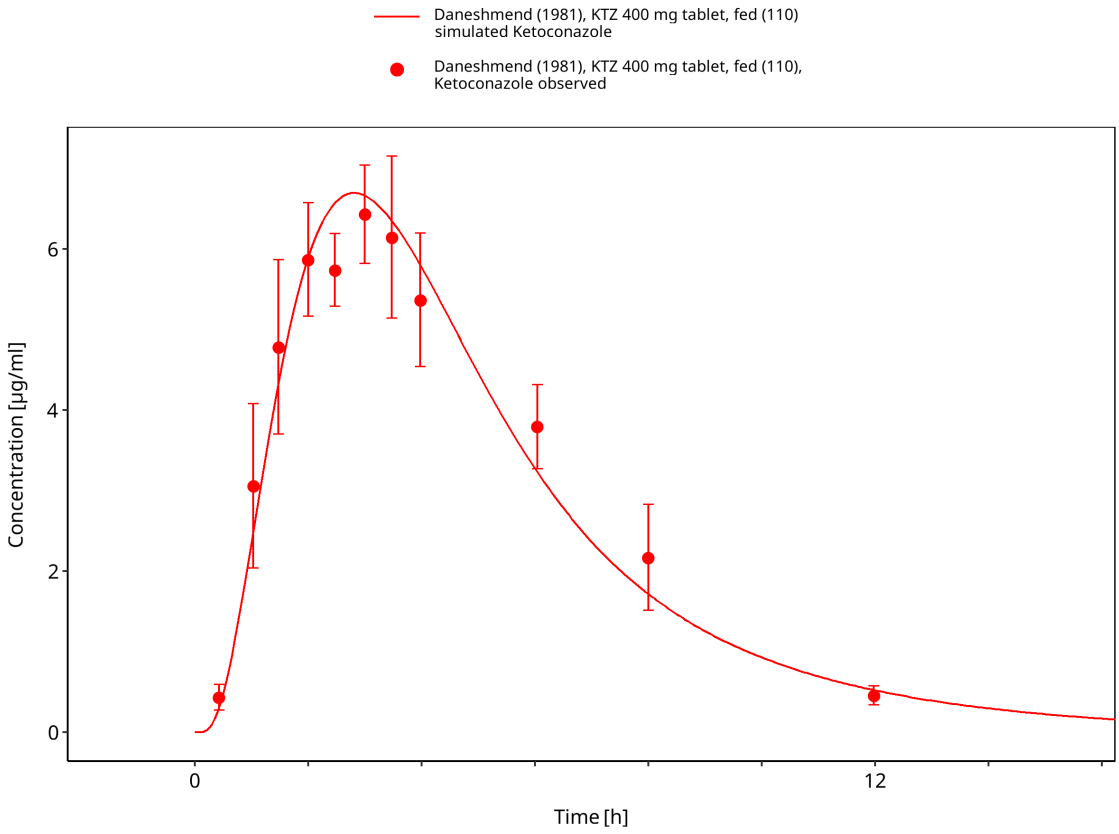


Figure 3-18: Daneshmend 1981: KTZ 400 mg tablet, fed (110)

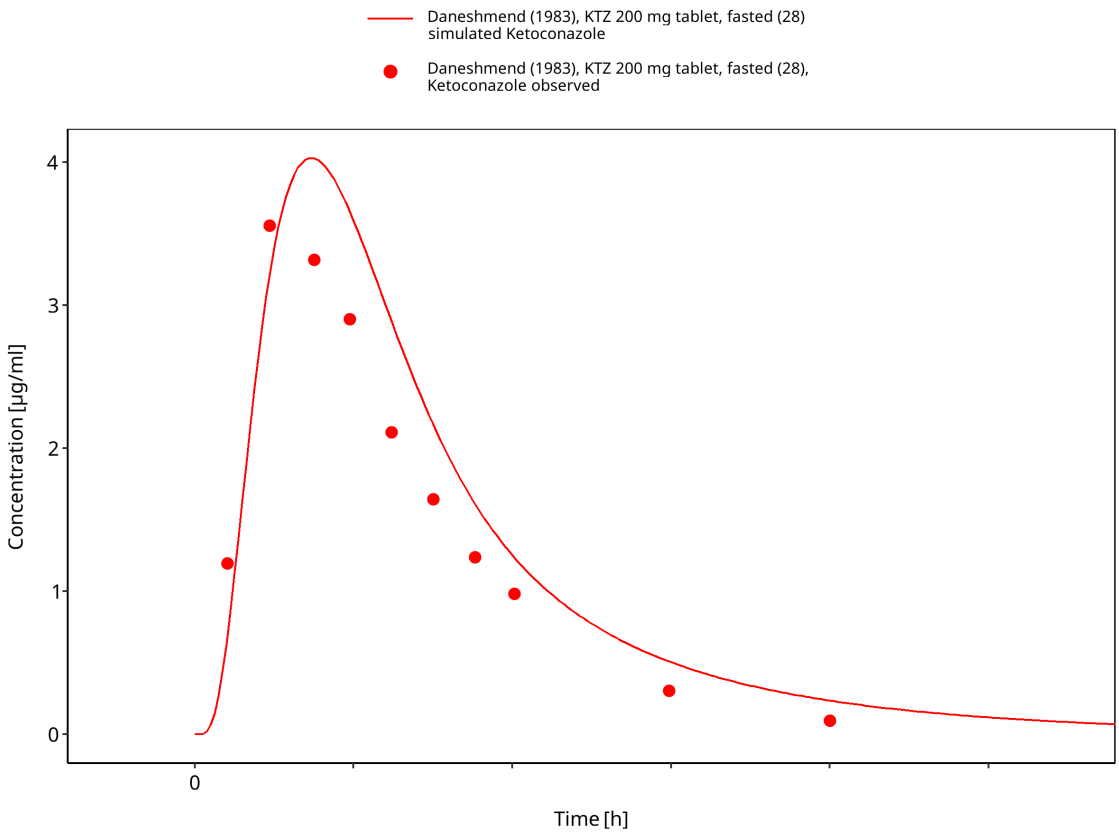


Figure 3-19: Daneshmend 1983: KTZ 200 mg tablet, fasted (28)

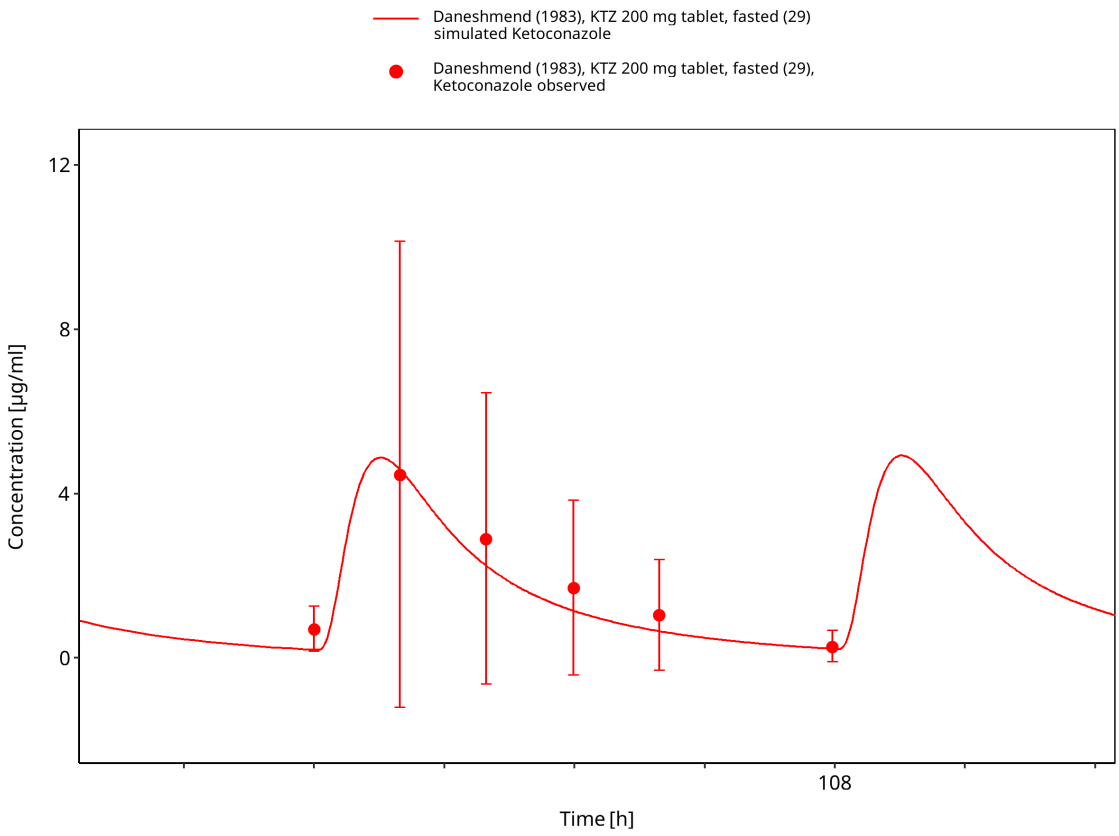


Figure 3-20: Daneshmend 1983: KTZ 200 mg tablet b.i.d. 5d, fasted (29)

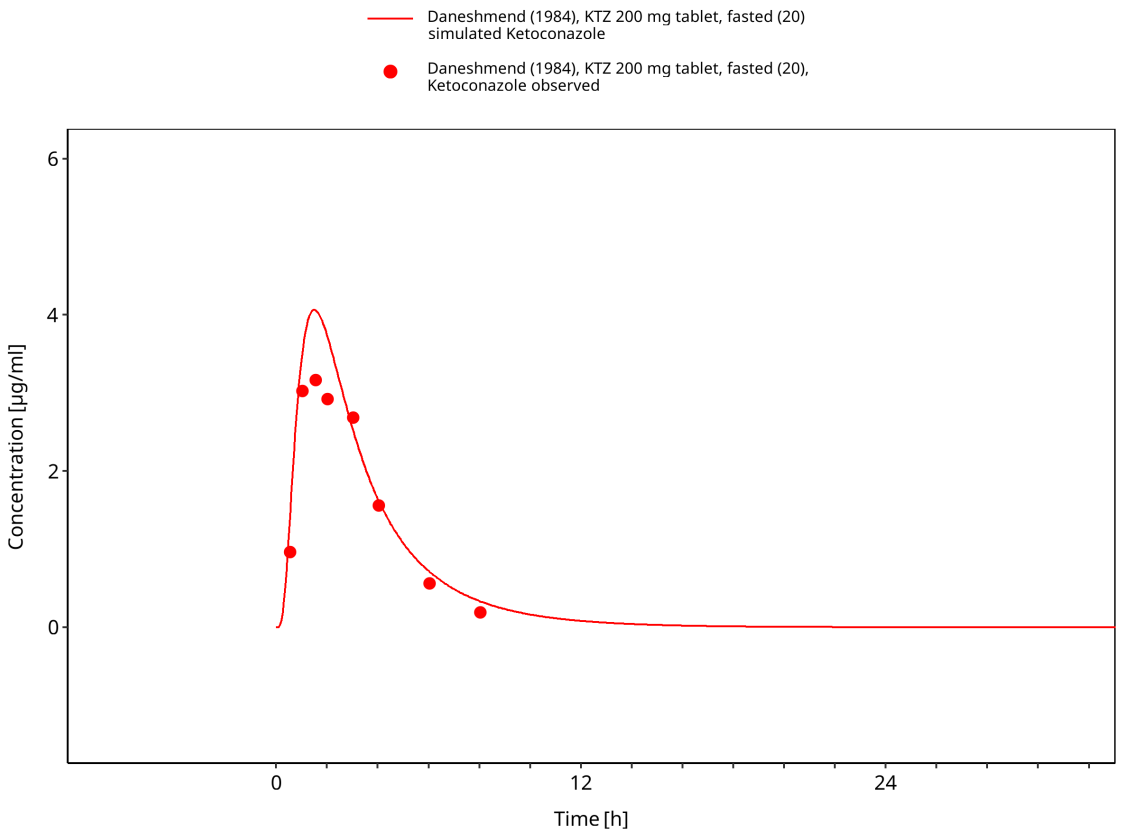


Figure 3-21: Daneshmend 1984: KTZ 200 mg tablet, fasted (20)

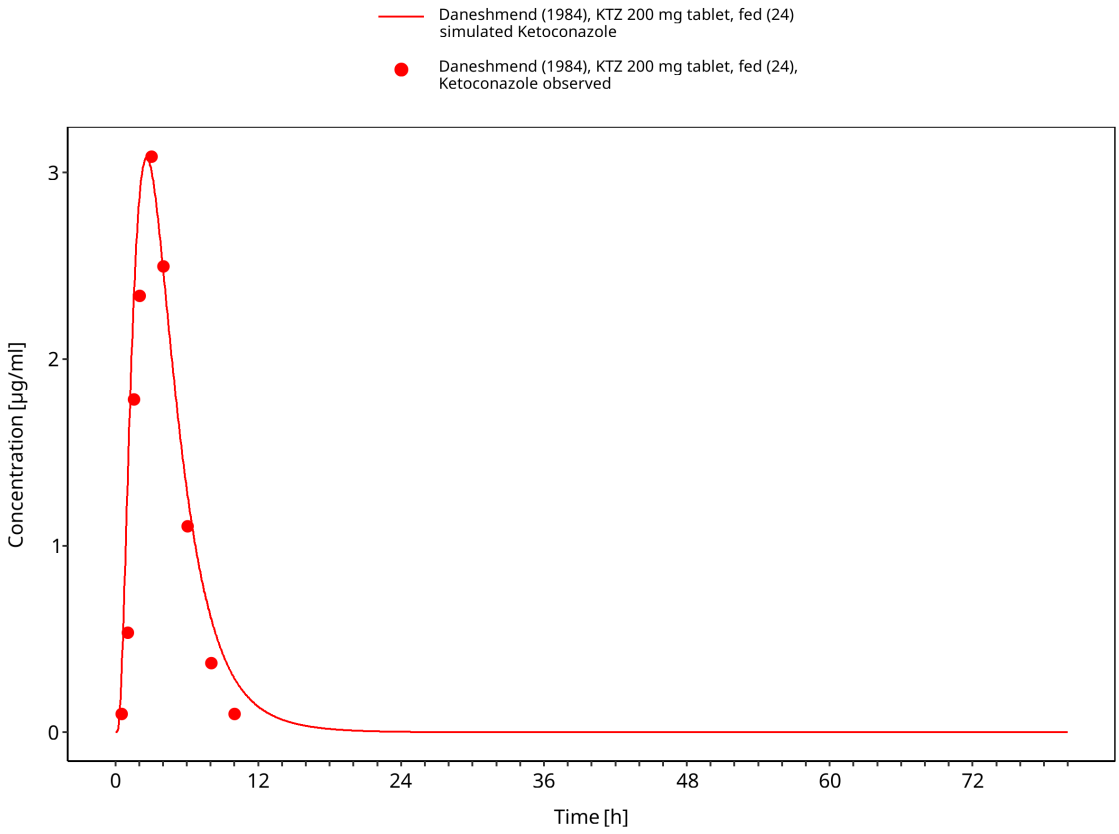


Figure 3-22: Daneshmend 1984: KTZ 200 mg tablet, fed (24)

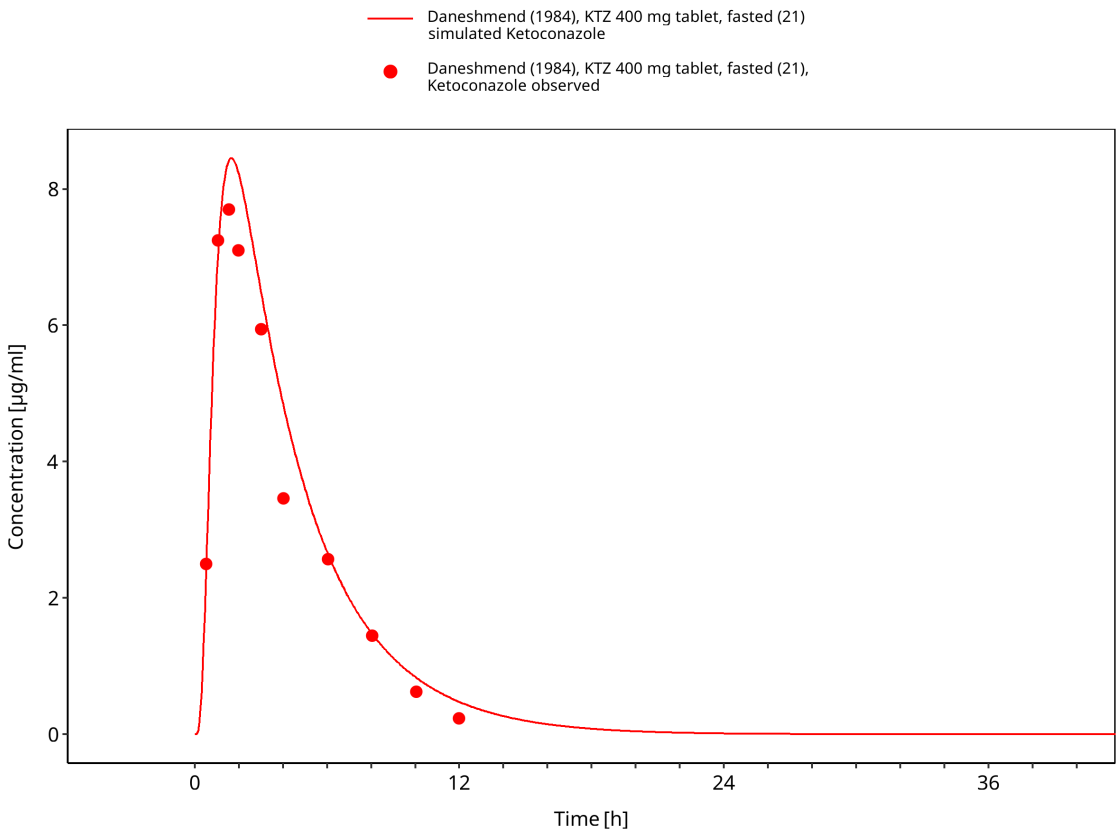


Figure 3-23: Daneshmend 1984: KTZ 400 mg tablet, fasted (21)

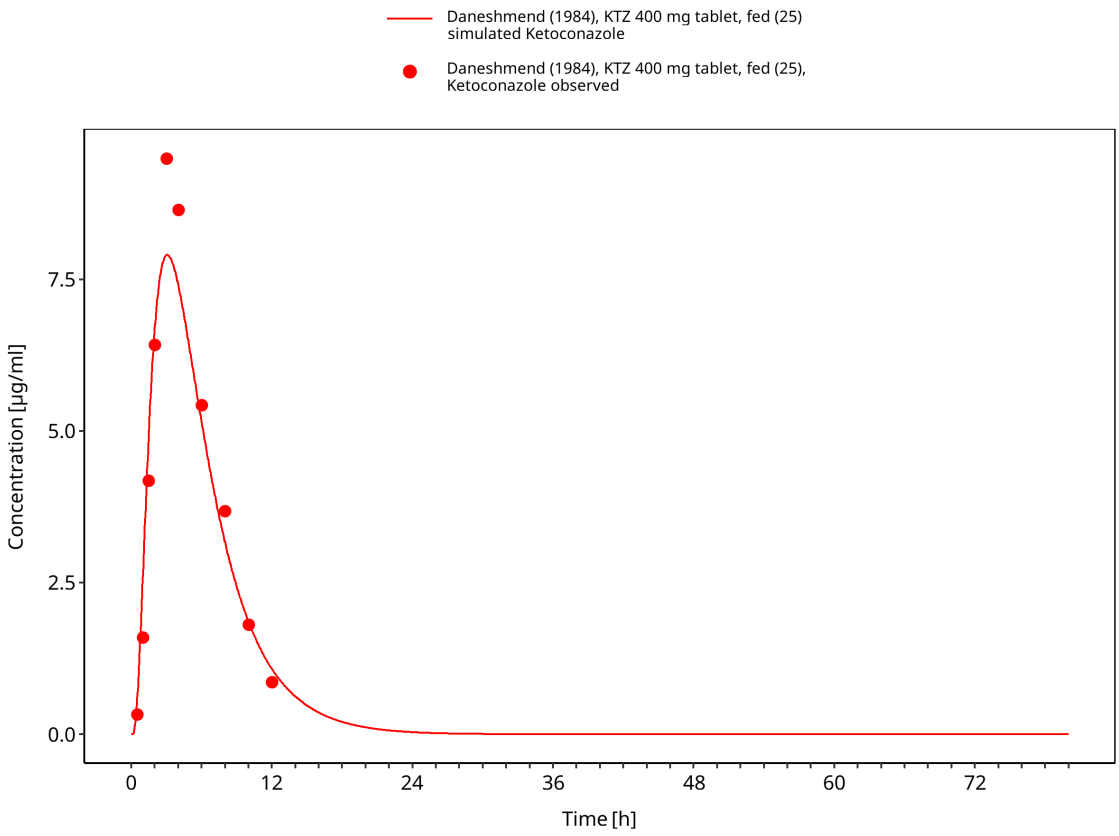


Figure 3-24: Daneshmend 1984: KTZ 400 mg tablet, fed (25)

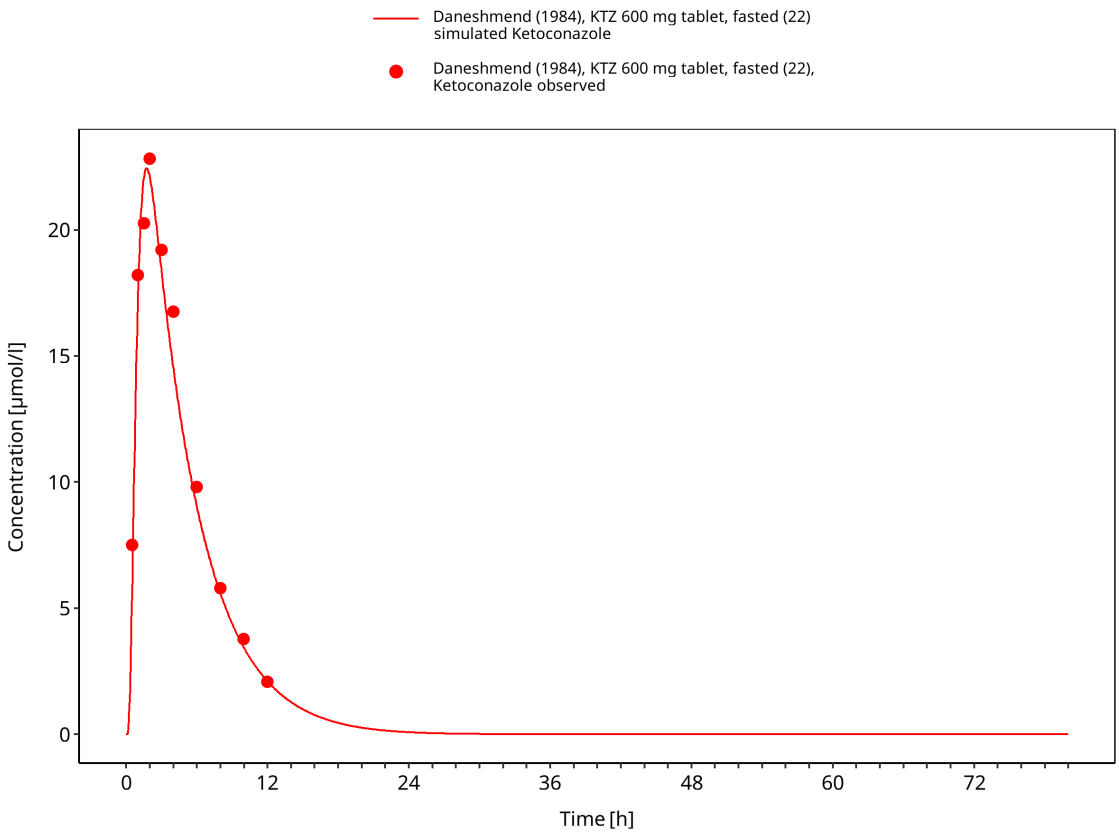


Figure 3-25: Daneshmend 1984: KTZ 600 mg tablet, fasted (22)

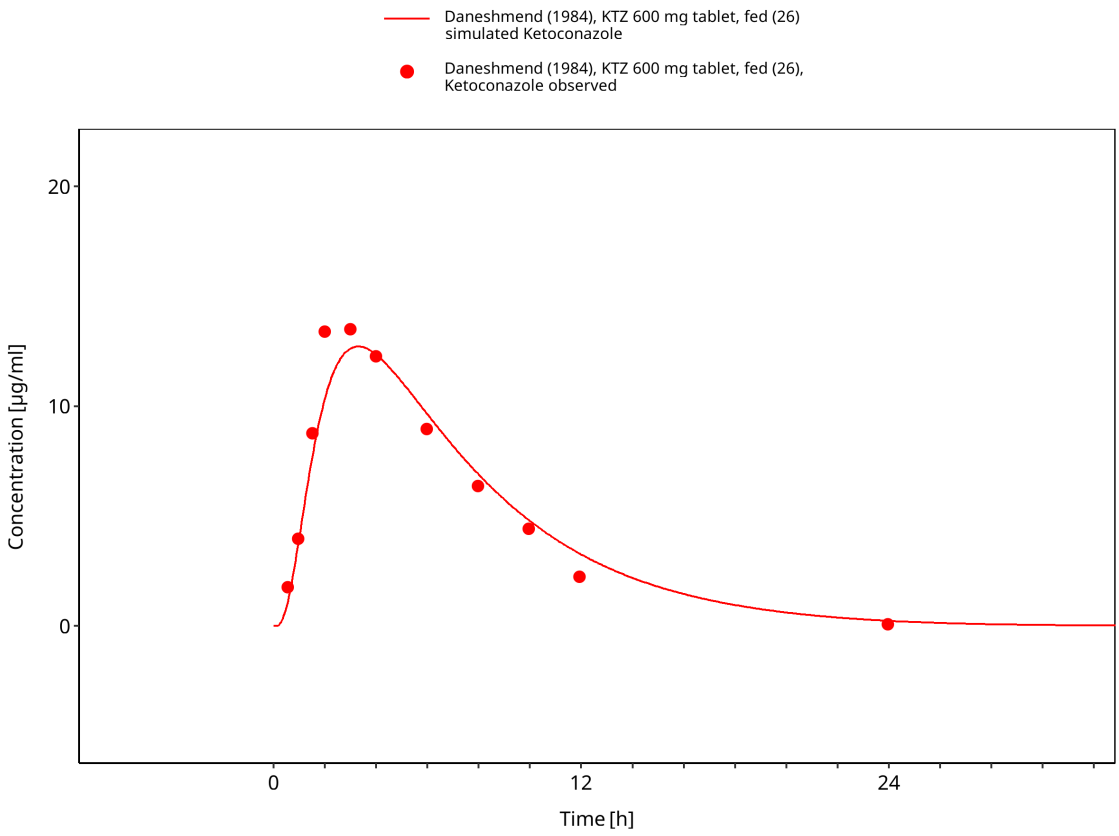


Figure 3-26: Daneshmend 1984: KTZ 600 mg tablet, fed (26)

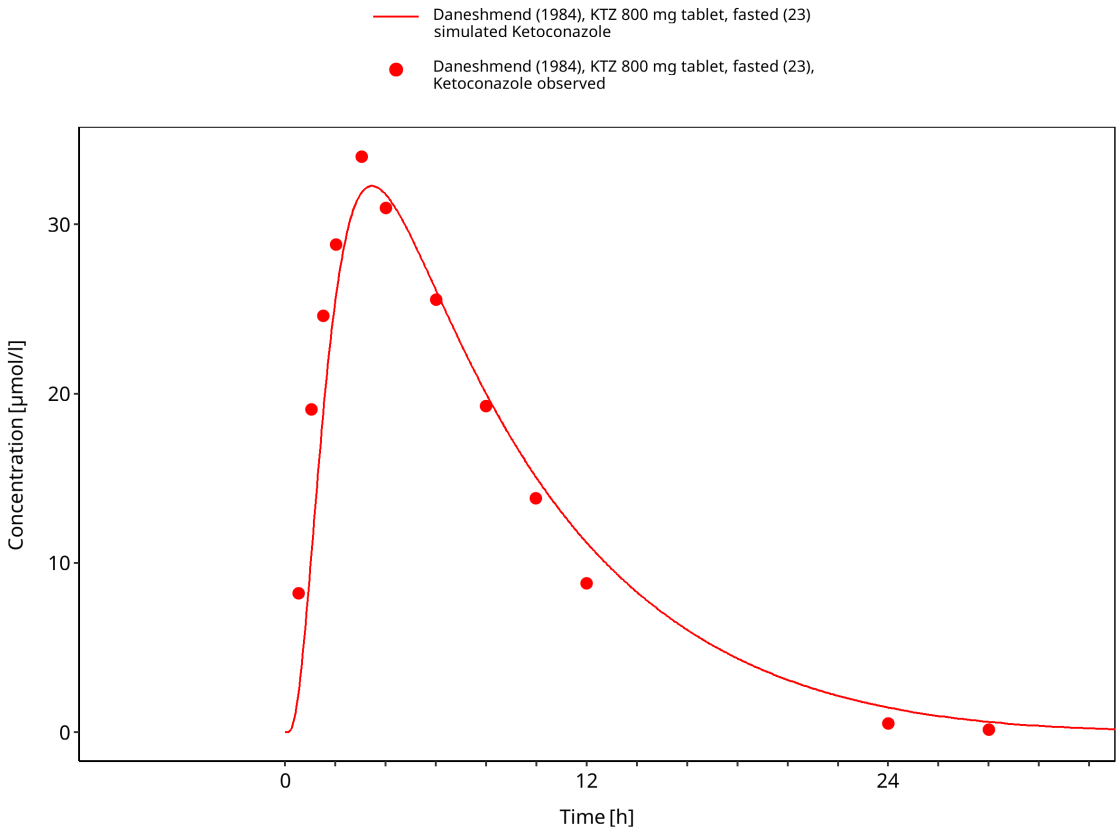


Figure 3-27: Daneshmend 1984: KTZ 800 mg tablet, fasted (23)

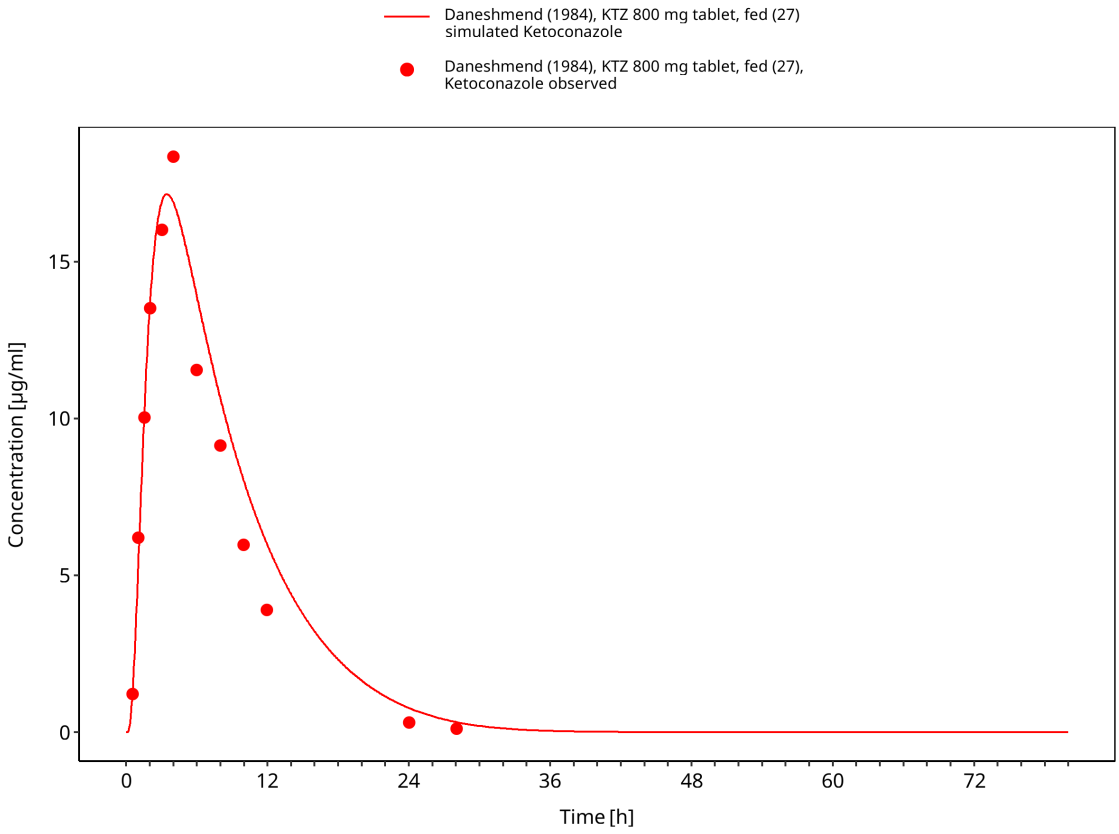


Figure 3-28: Daneshmend 1984: KTZ 800 mg tablet, fed (27)

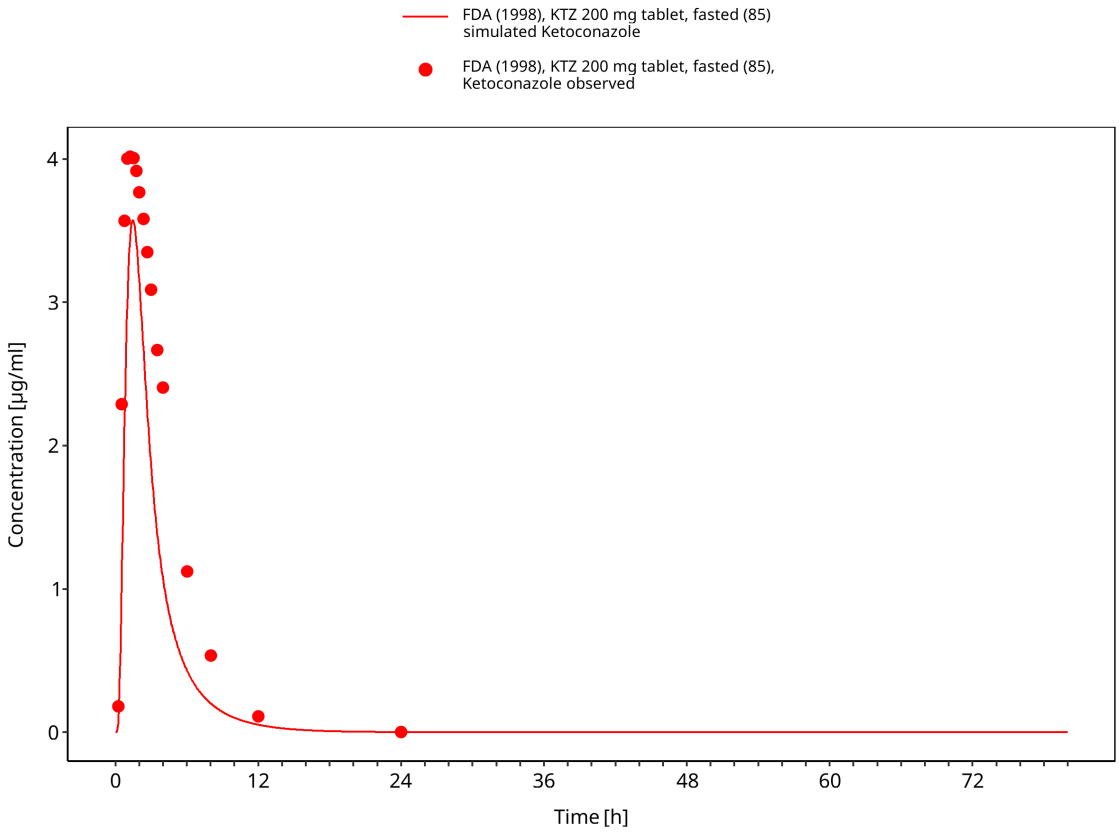


Figure 3-29: FDA 1998: KTZ 200 mg tablet, fasted (85)

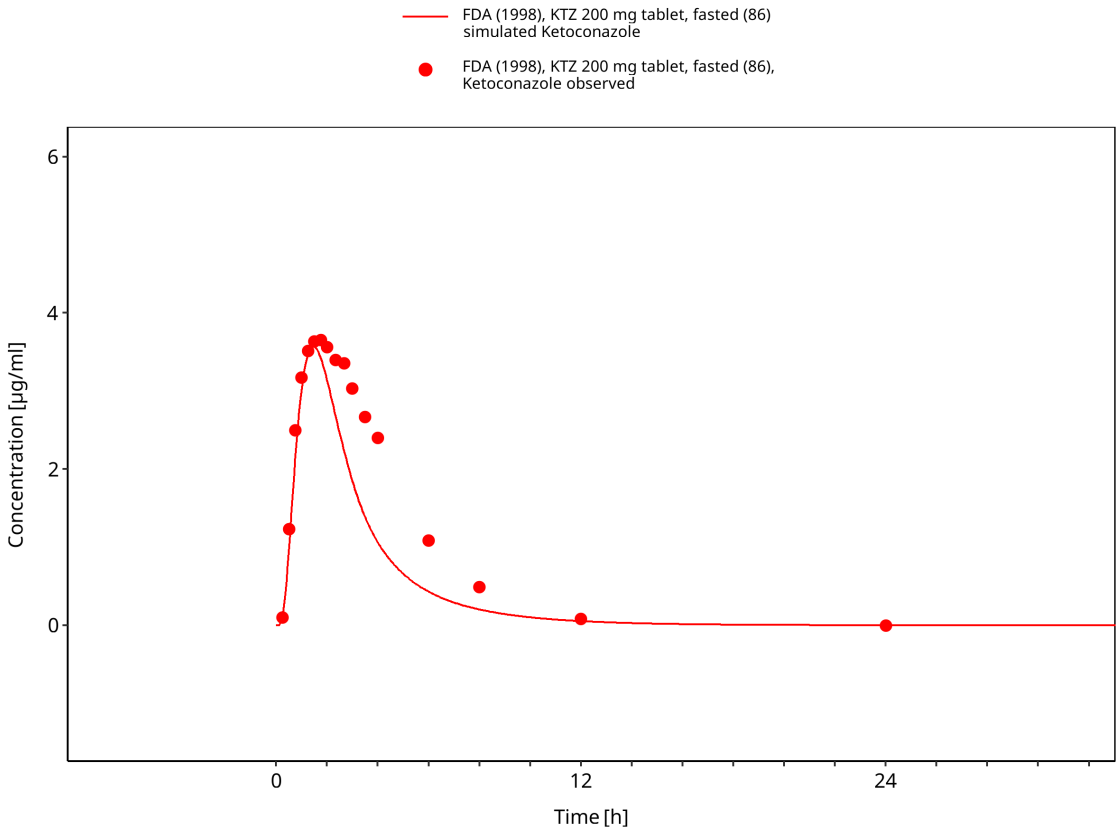


Figure 3-30: FDA 1998: KTZ 200 mg tablet, fasted (86)

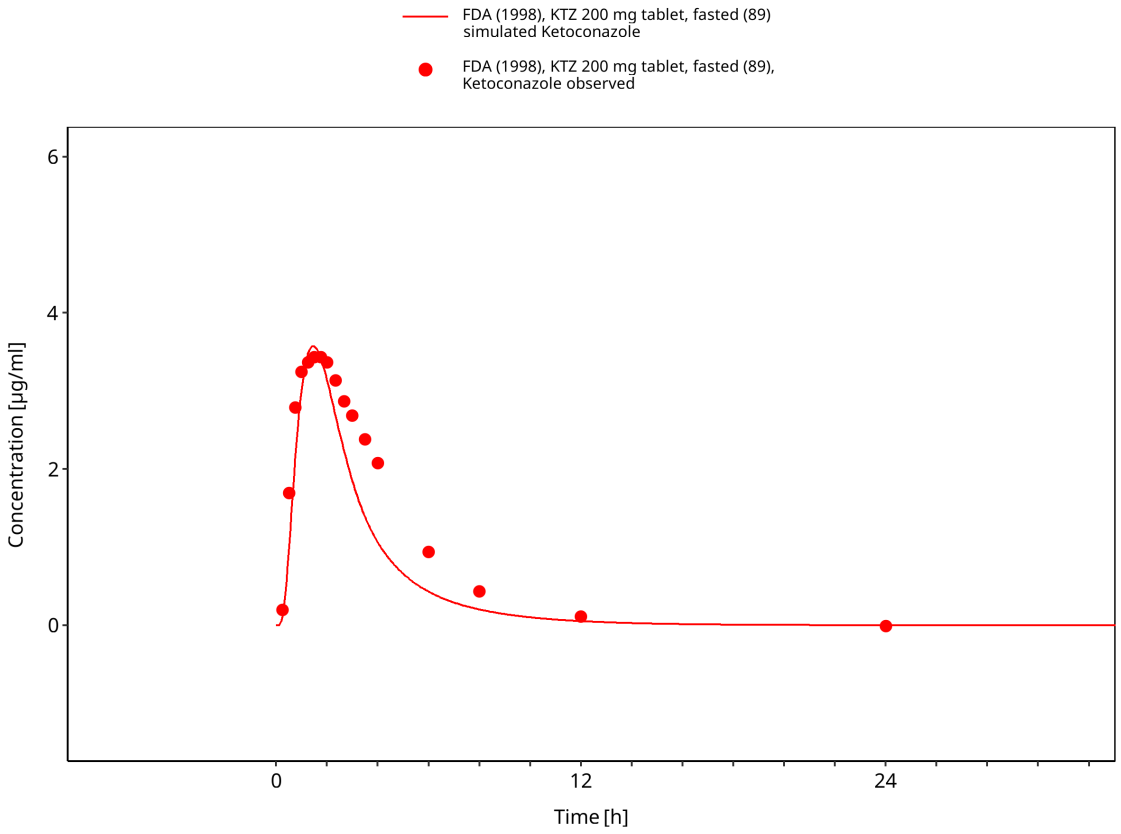


Figure 3-31: FDA 1998: KTZ 200 mg tablet, fasted (89)

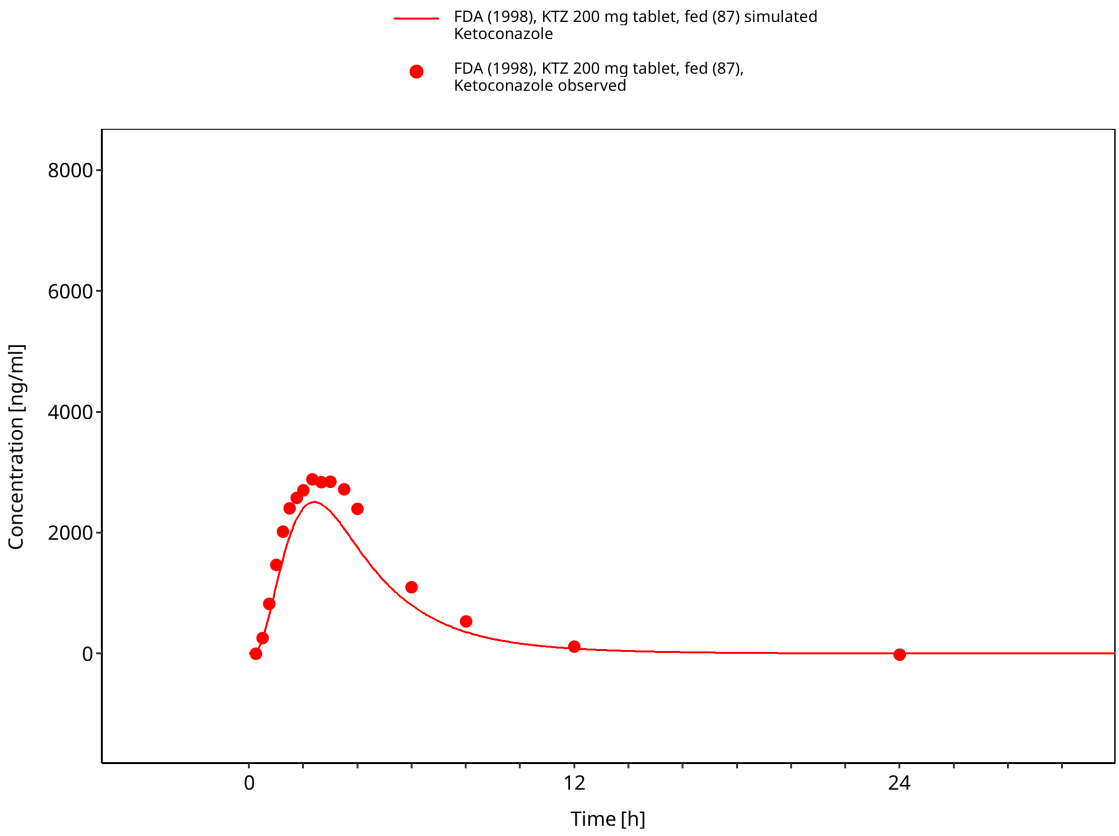


Figure 3-32: FDA 1998: KTZ 200 mg tablet, fed (87)

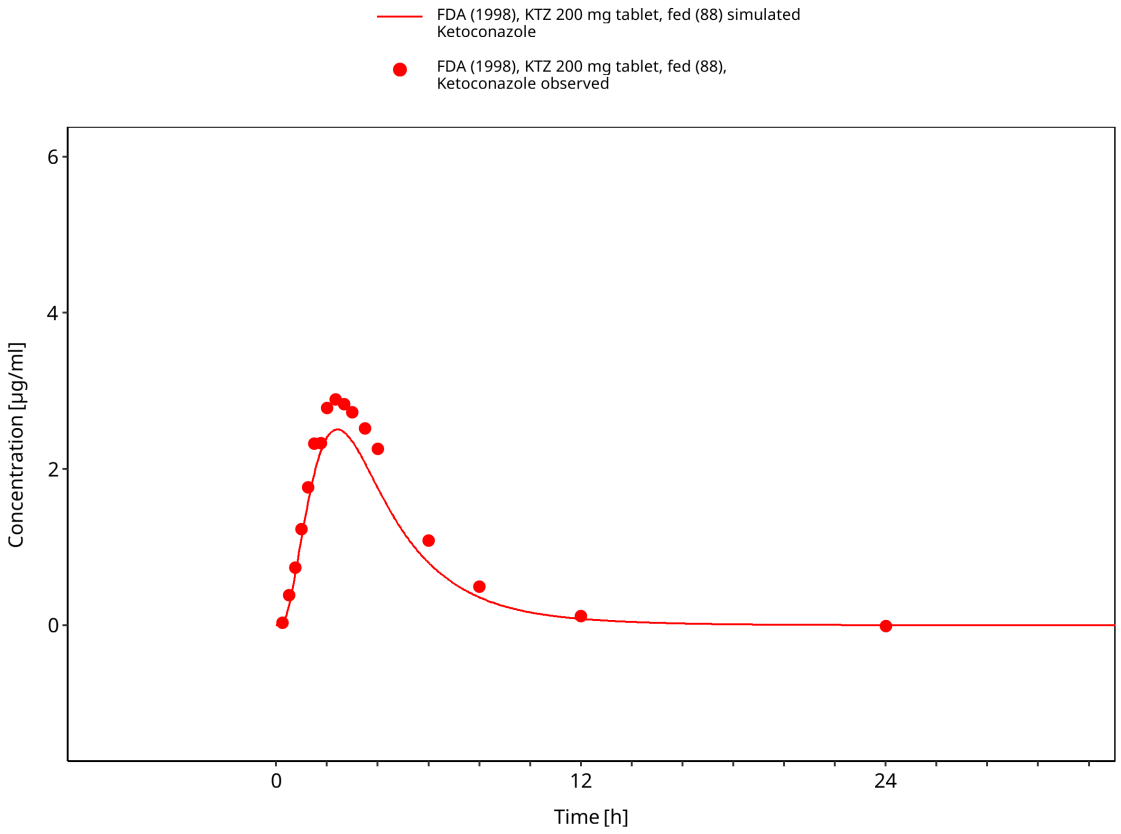


Figure 3-33: FDA 1998: KTZ 200 mg tablet, fed (88)

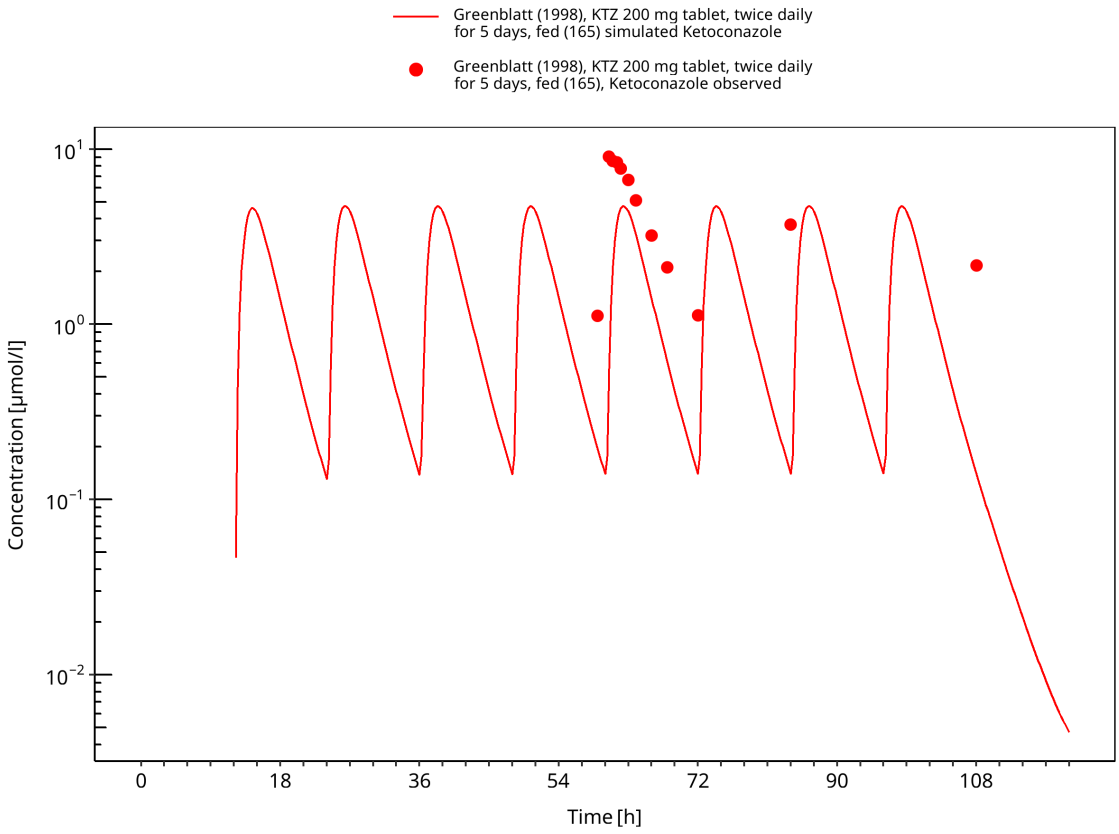


Figure 3-34: Greenblatt 1998: KTZ 200 mg tablet b.i.d. 5d, fed (165)

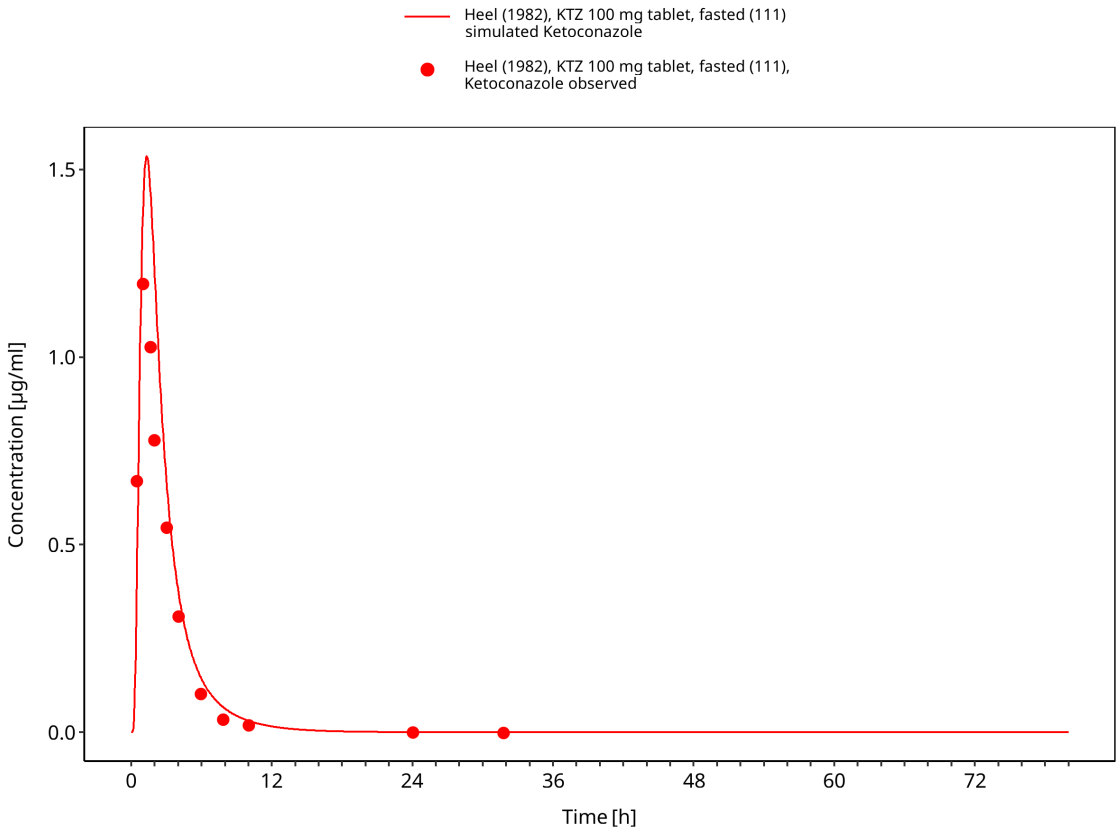


Figure 3-35: Heel 1982: KTZ 100 mg tablet, fasted (111)

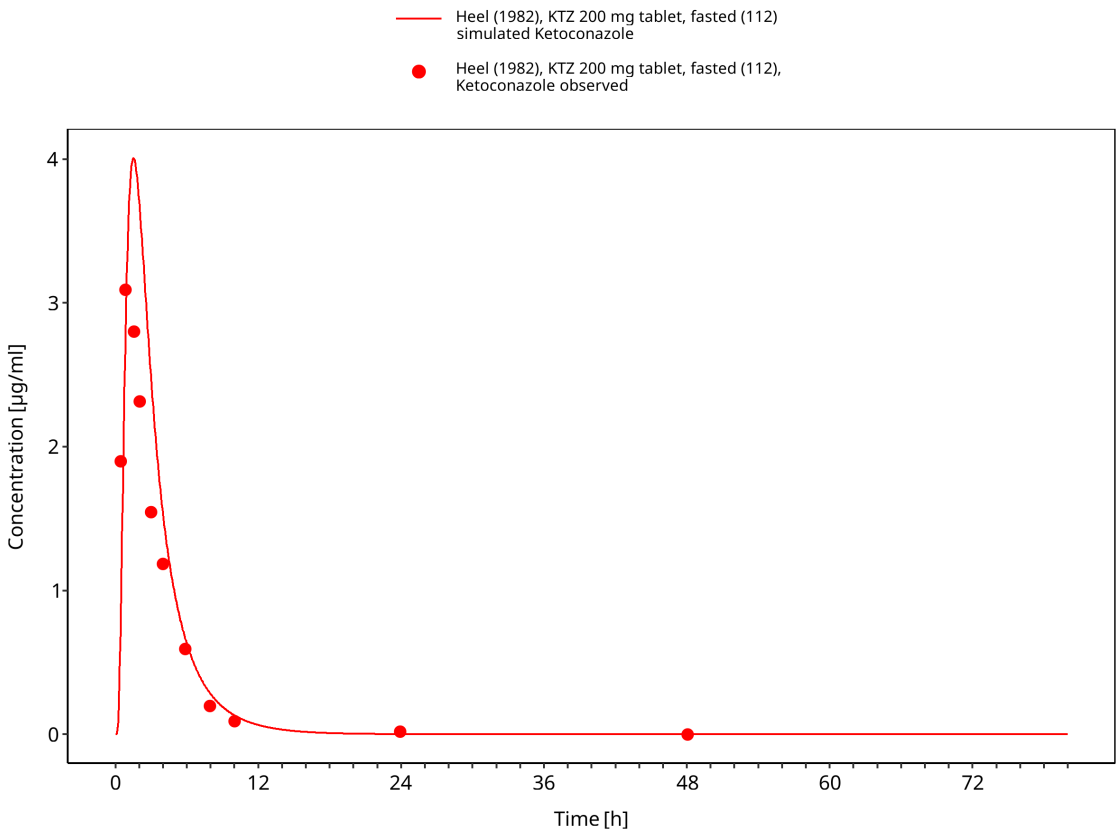


Figure 3-36: Heel 1982: KTZ 200 mg tablet, fasted (112)

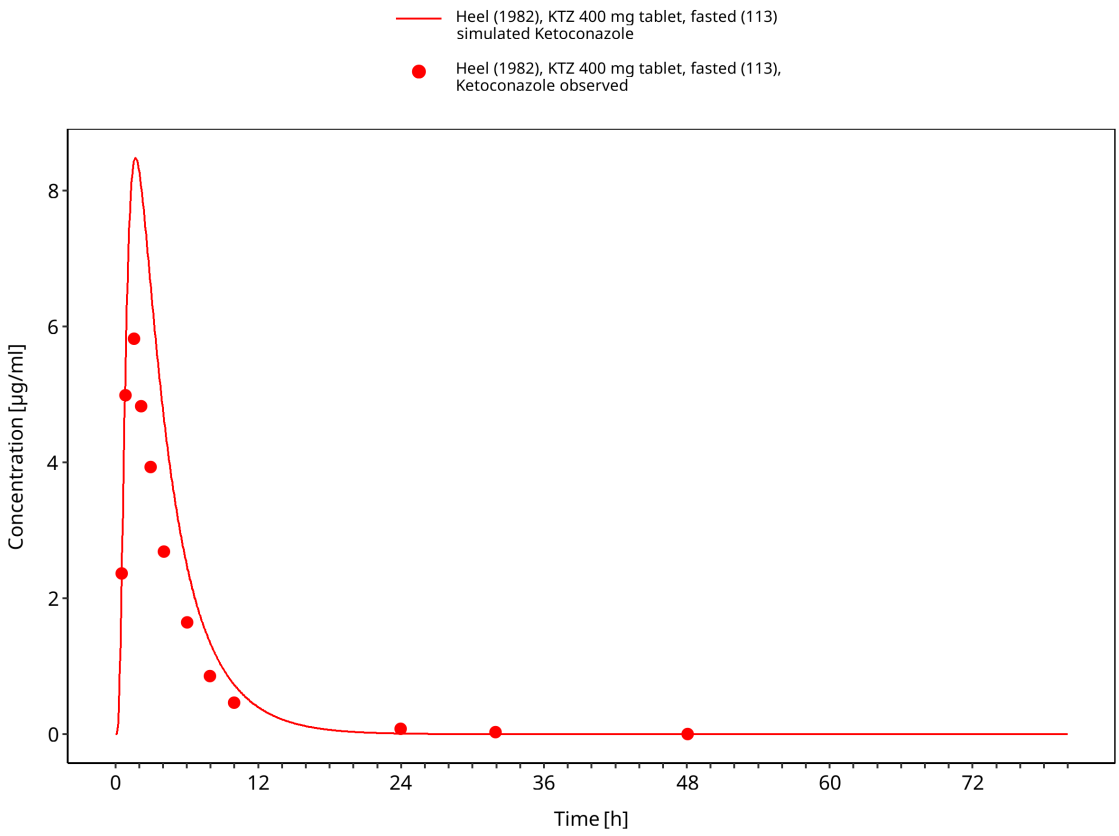


Figure 3-37: Heel 1982: KTZ 400 mg tablet, fasted (113)

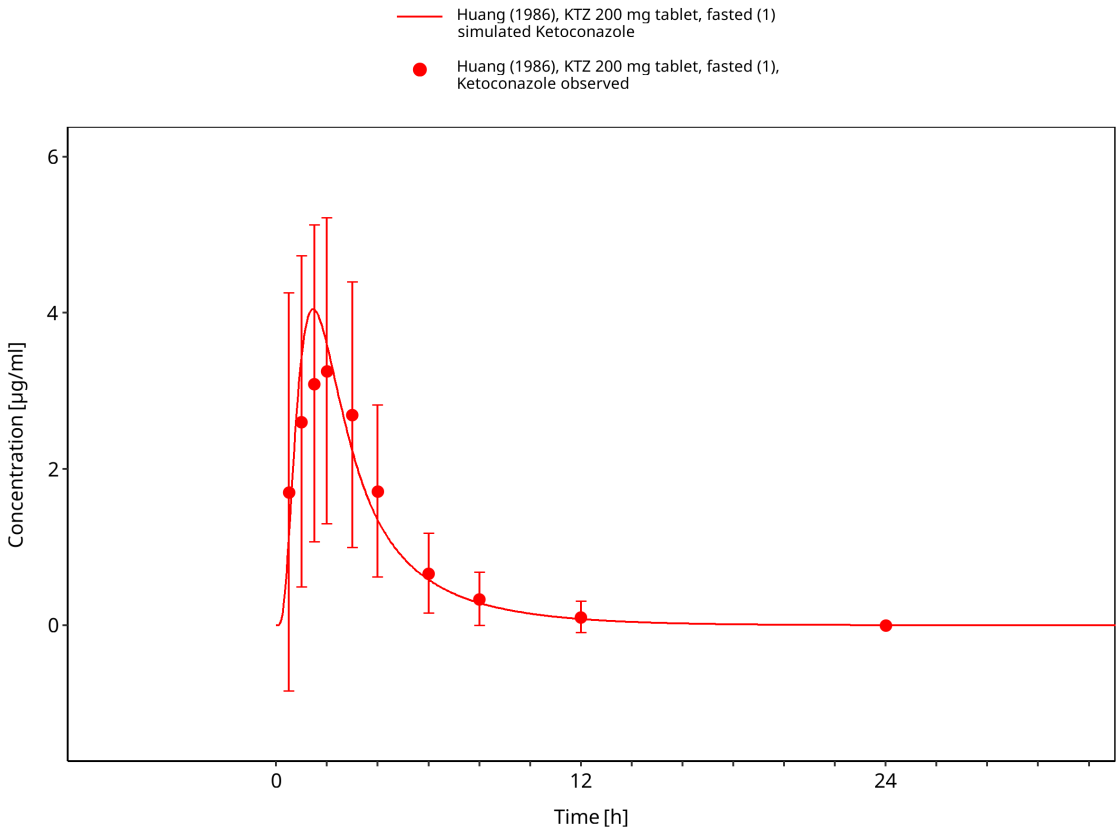


Figure 3-38: Huang 1986: KTZ 200 mg tablet, fasted (1)

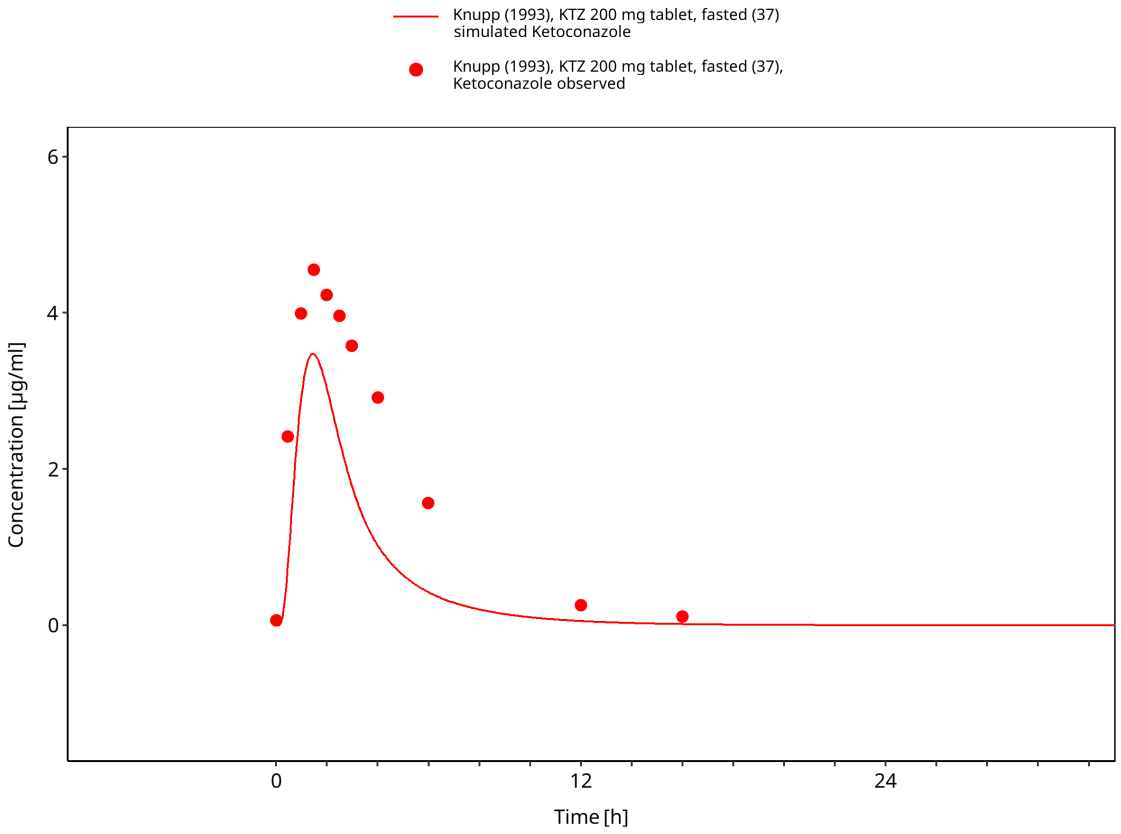


Figure 3-39: Knupp 1993: KTZ 200 mg tablet, fasted (37)

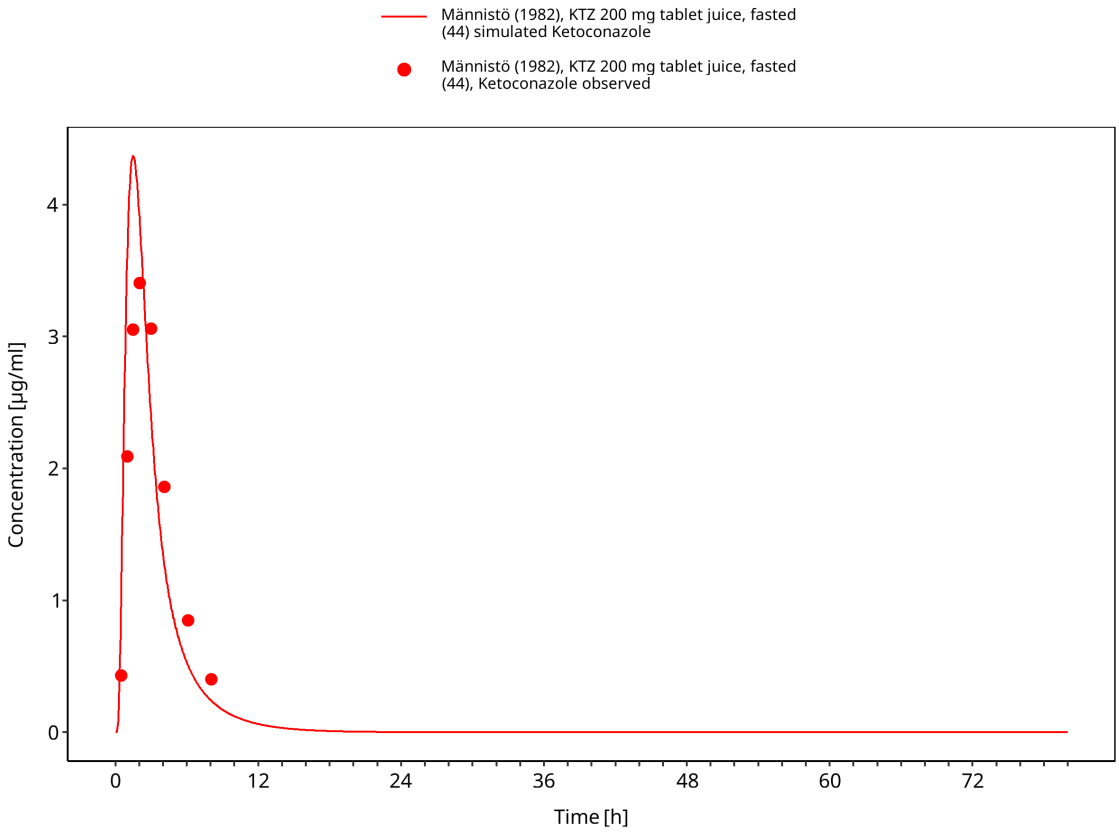


Figure 3-40: Männistö 1982: KTZ 200 mg tablet juice, fasted (44)

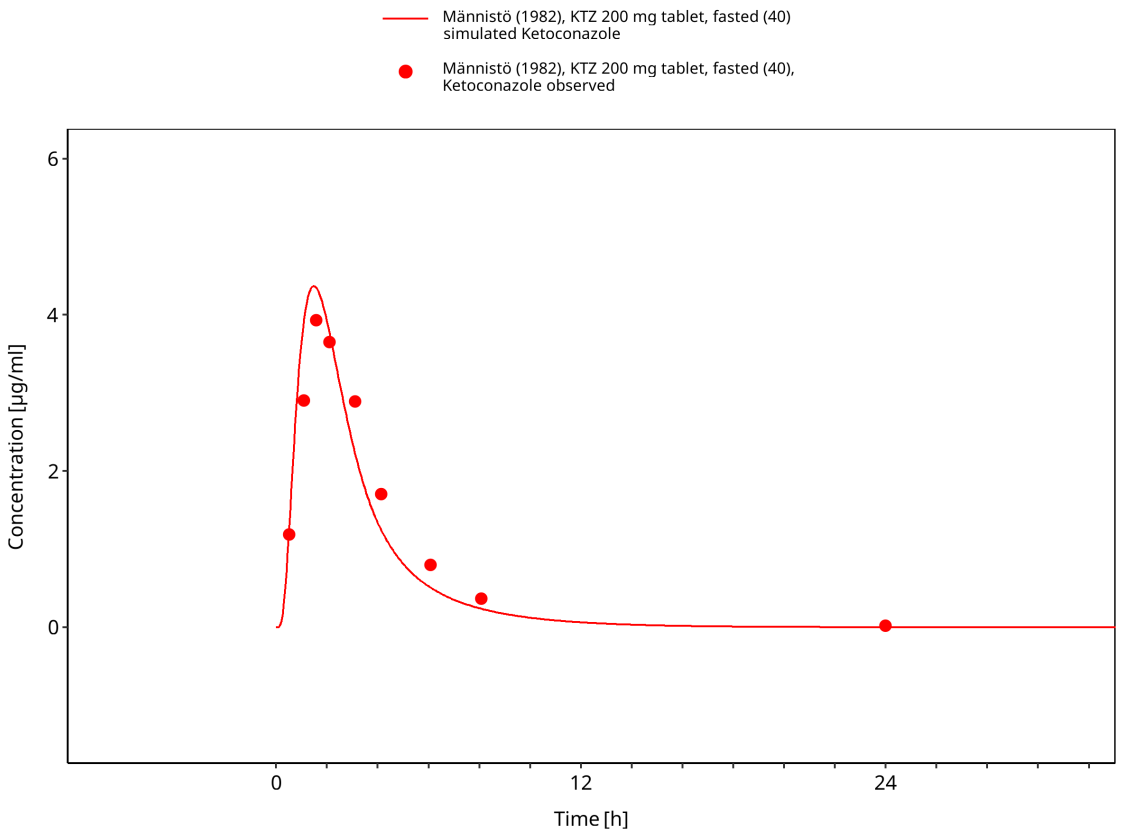


Figure 3-41: Männistö 1982: KTZ 200 mg tablet, fasted (40)

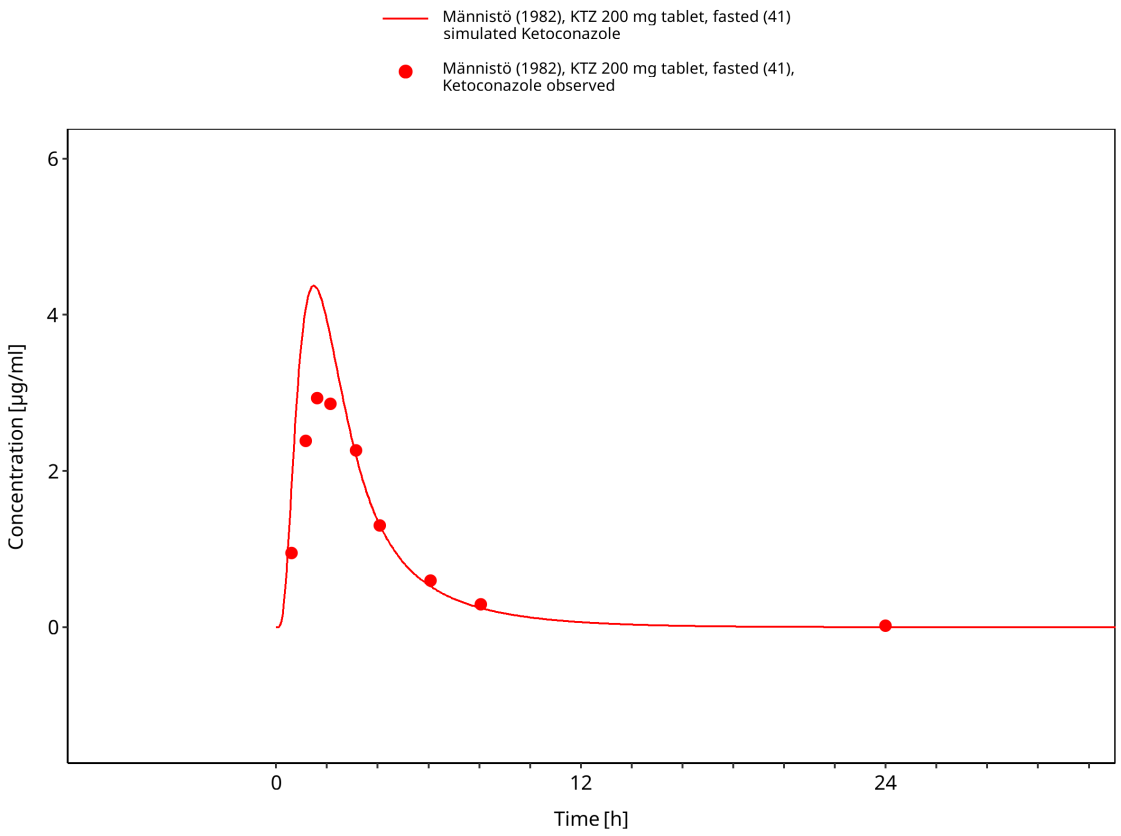


Figure 3-42: Männistö 1982: KTZ 200 mg tablet, fasted (41)

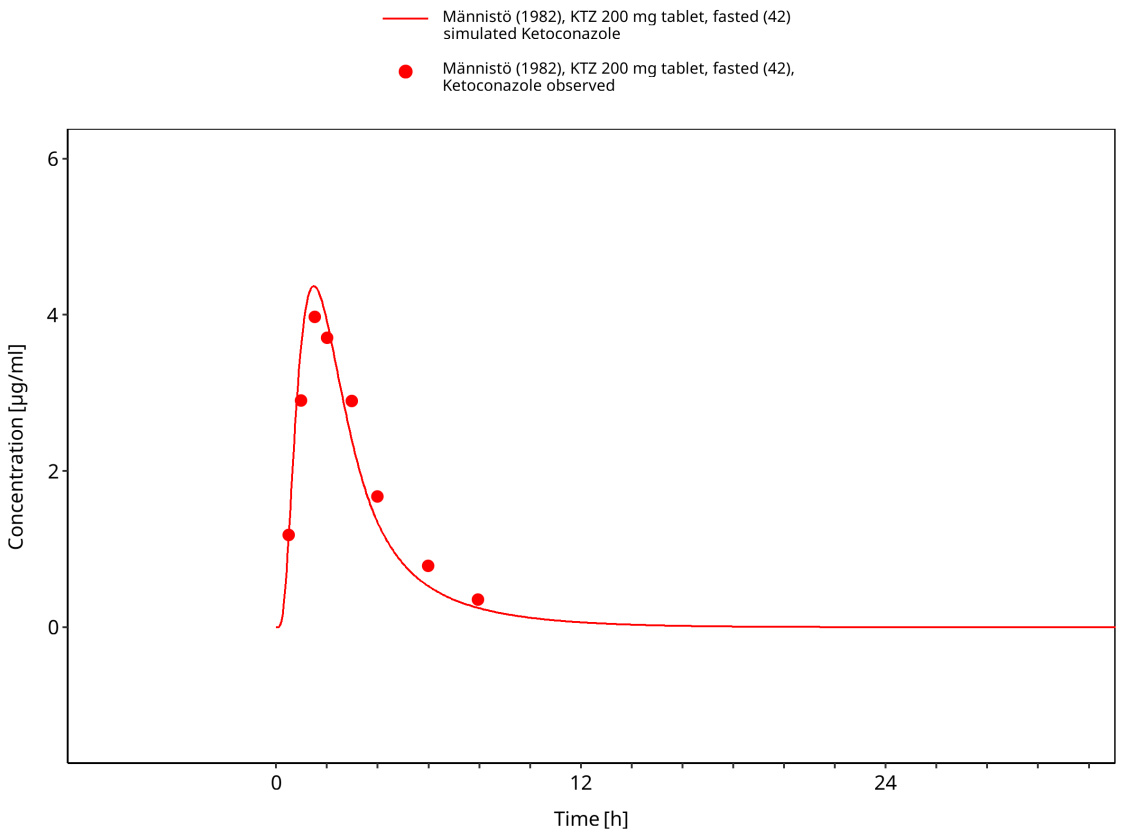


Figure 3-43: Männistö 1982: KTZ 200 mg tablet, fasted (42)

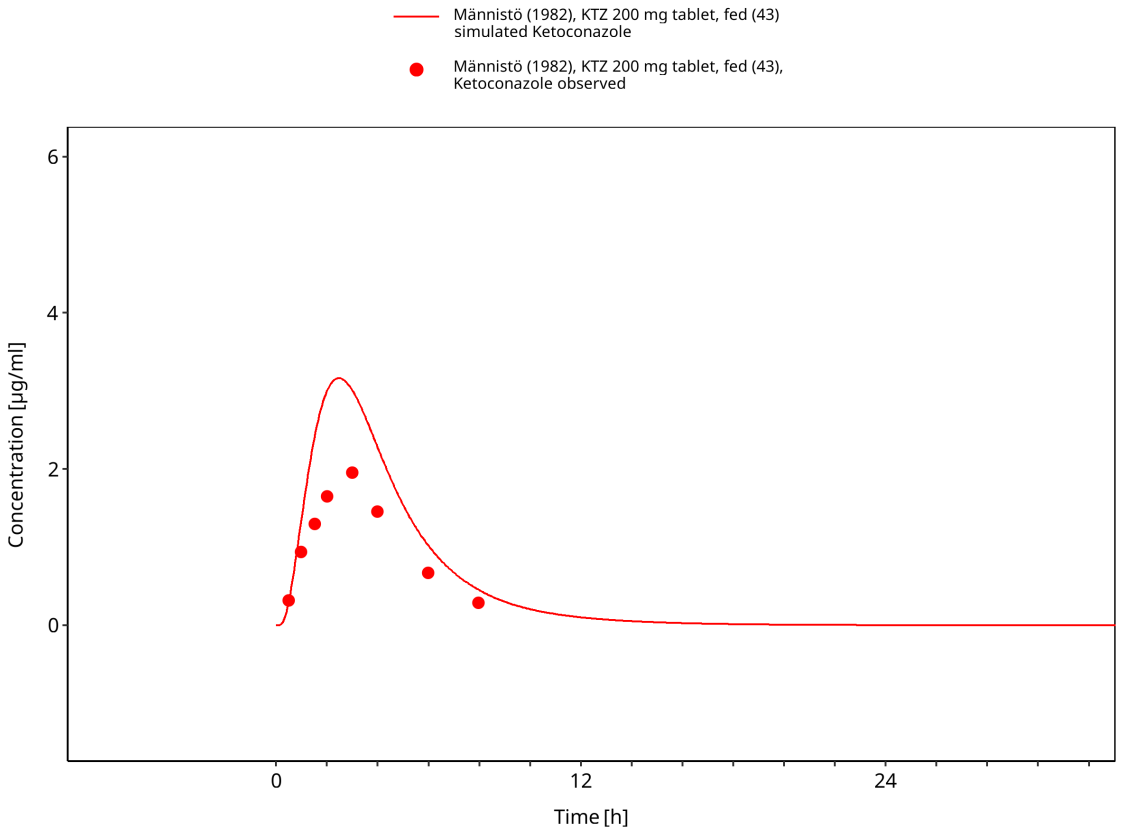


Figure 3-44: Männistö 1982: KTZ 200 mg tablet, fed (43)

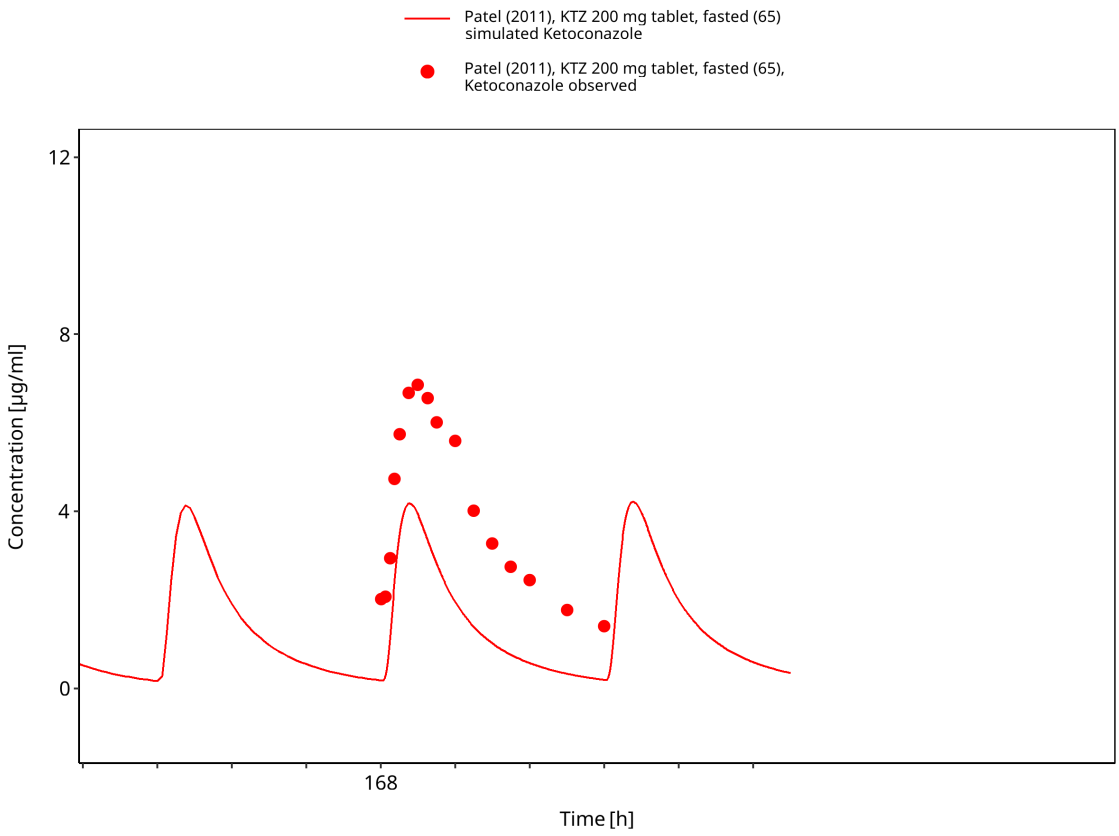


Figure 3-45: Patel 2011: KTZ 200 mg tablet b.i.d. 6d, fasted (65)

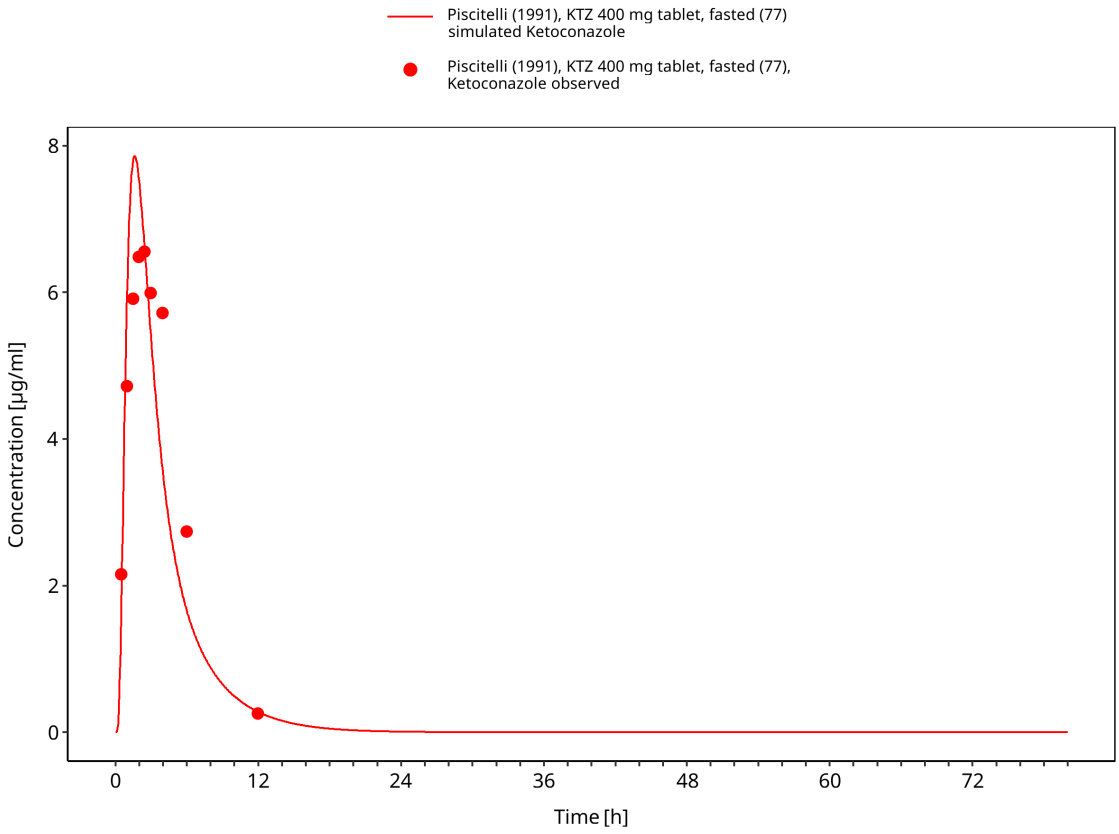


Figure 3-46: Piscitelli 1991: KTZ 400 mg tablet, fasted (77)

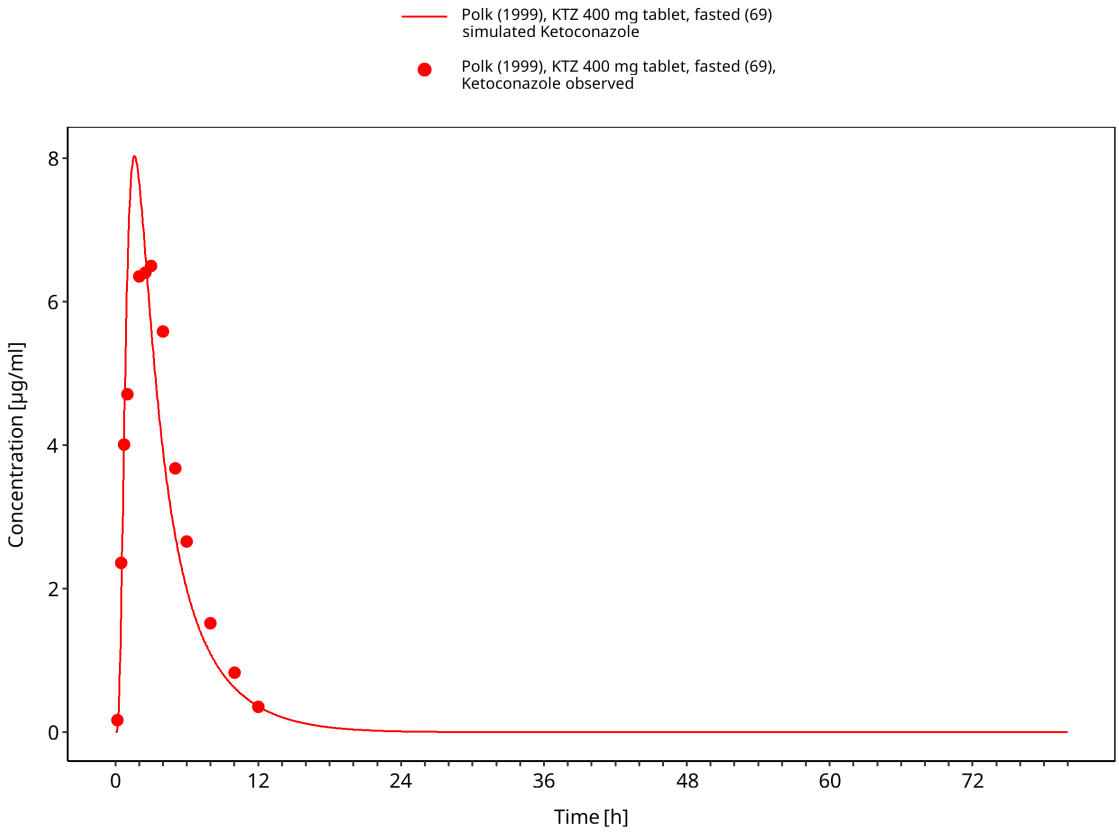


Figure 3-47: Polk 1999: KTZ 400 mg tablet, fasted (69)

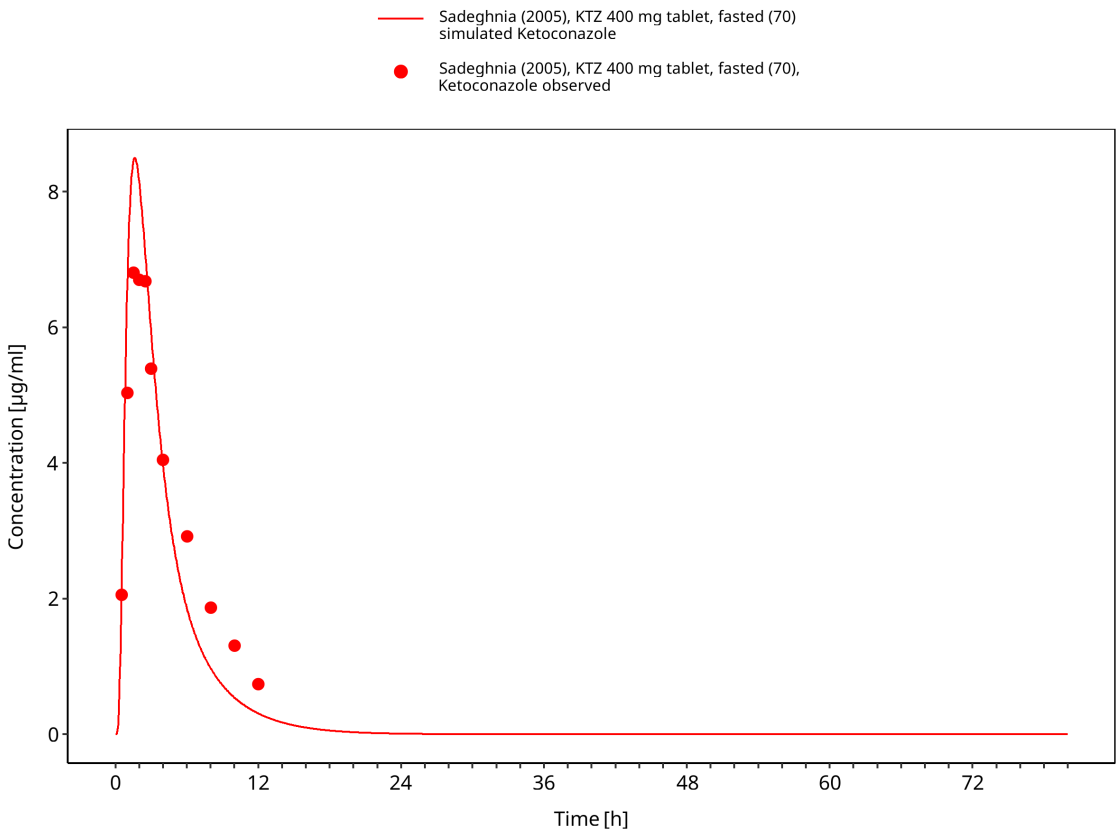


Figure 3-48: Sadeghnia 2005: KTZ 400 mg tablet, fasted (70)

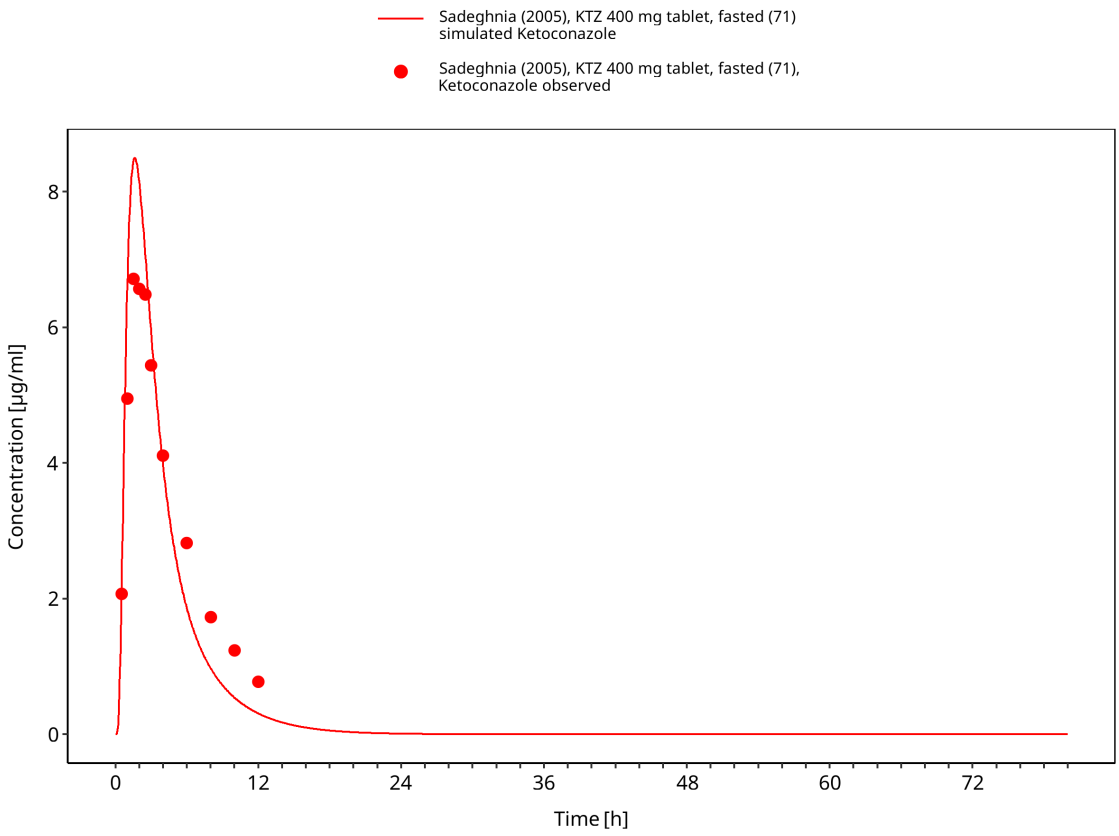


Figure 3-49: Sadeghnia 2005: KTZ 400 mg tablet, fasted (71)

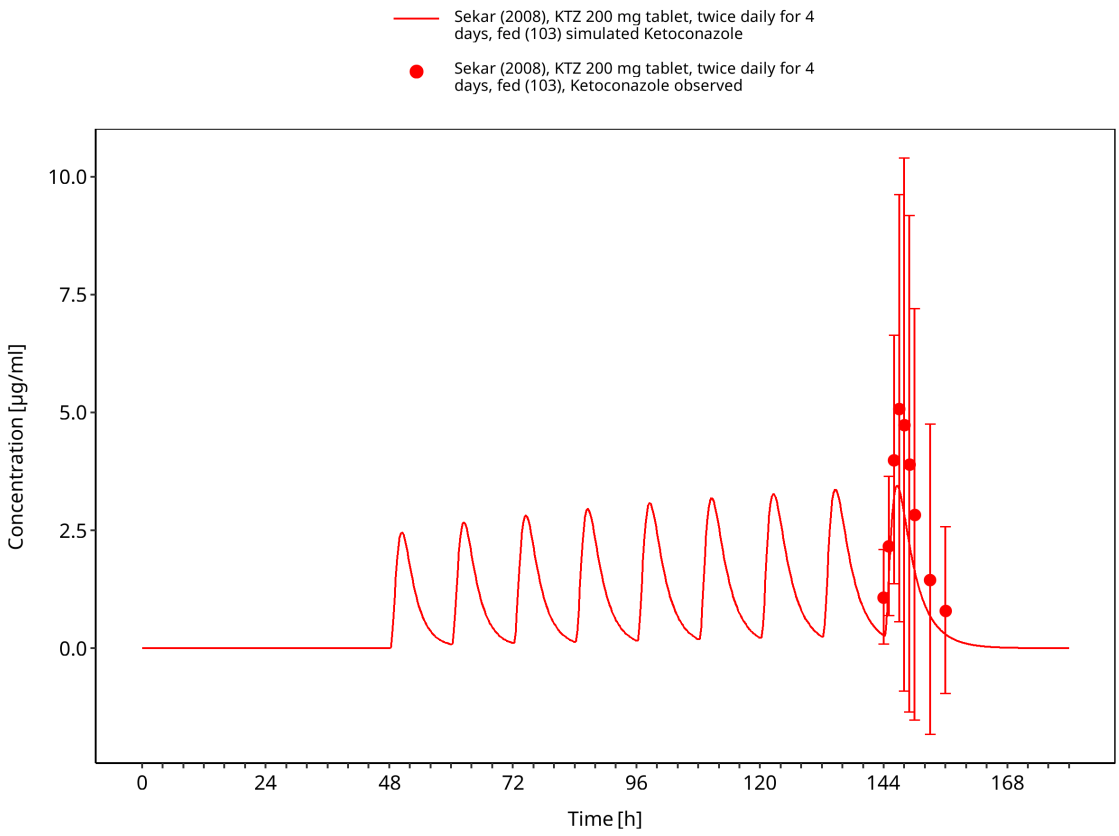


Figure 3-50: Sekar 2008: KTZ 200 mg tablet b.i.d., fed (103)

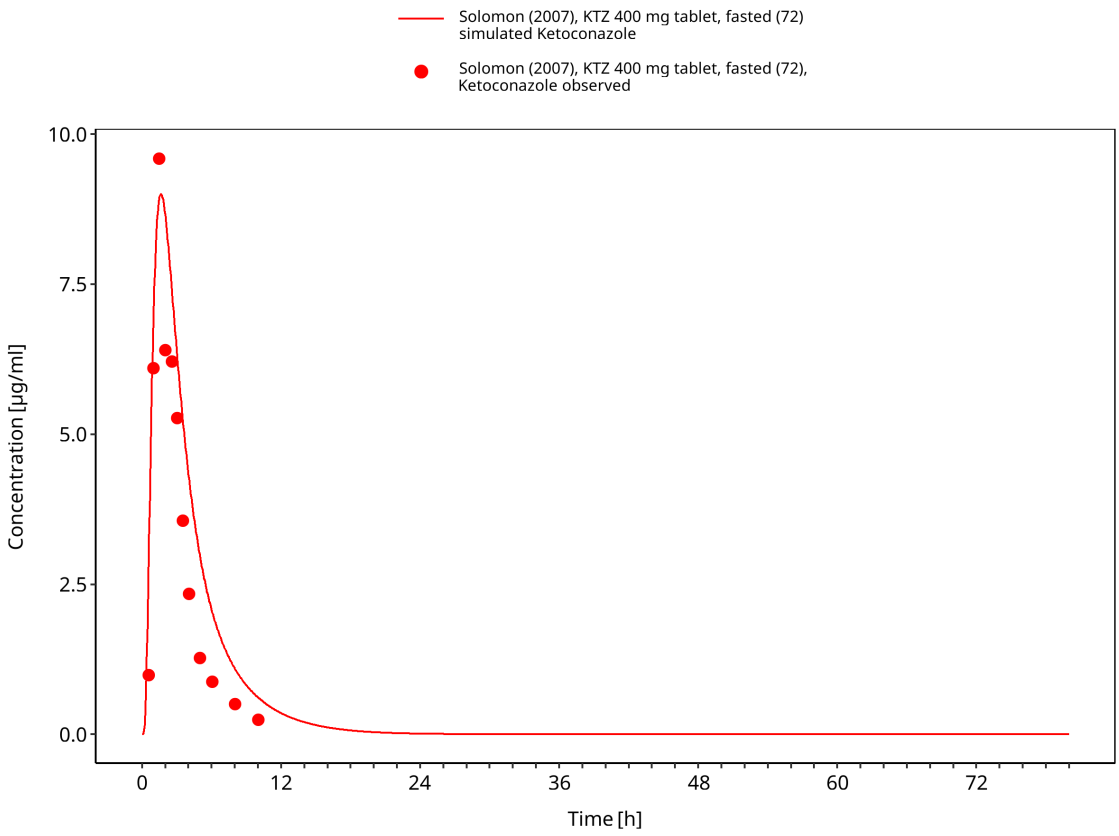


Figure 3-51: Solomon 2007: KTZ 400 mg tablet, fasted (72)

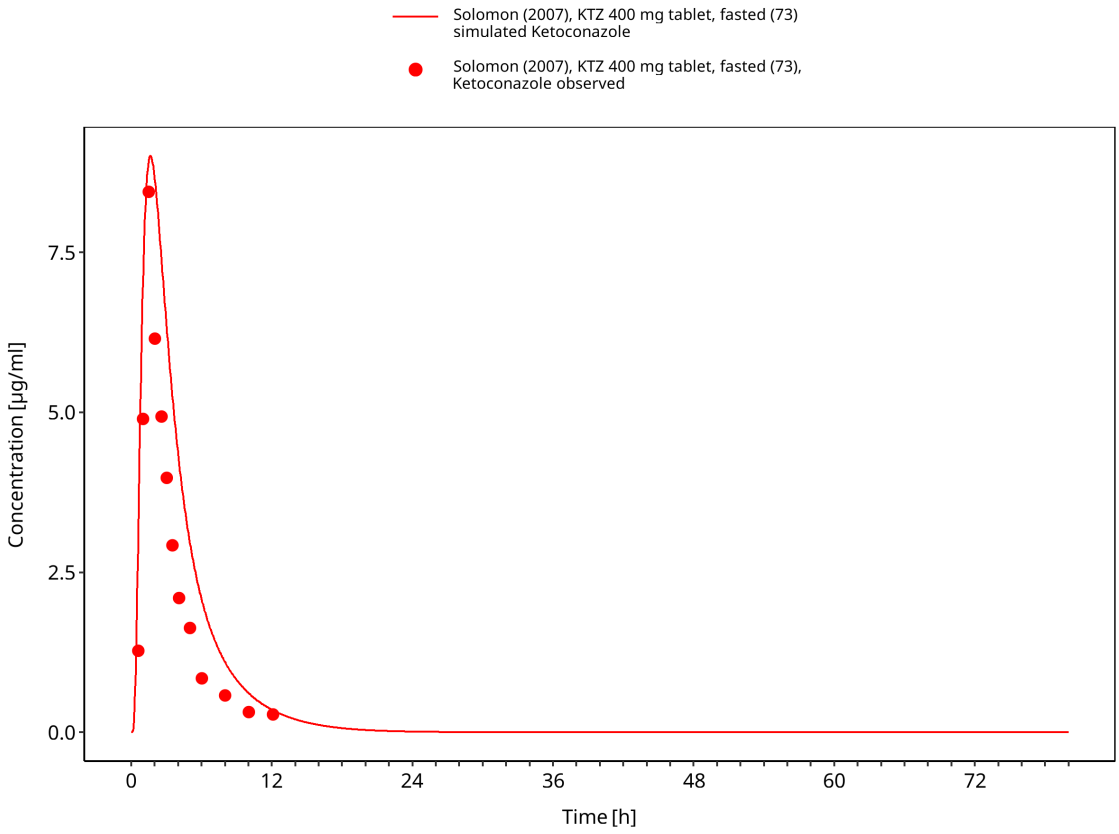


Figure 3-52: Solomon 2007: KTZ 400 mg tablet, fasted (73)

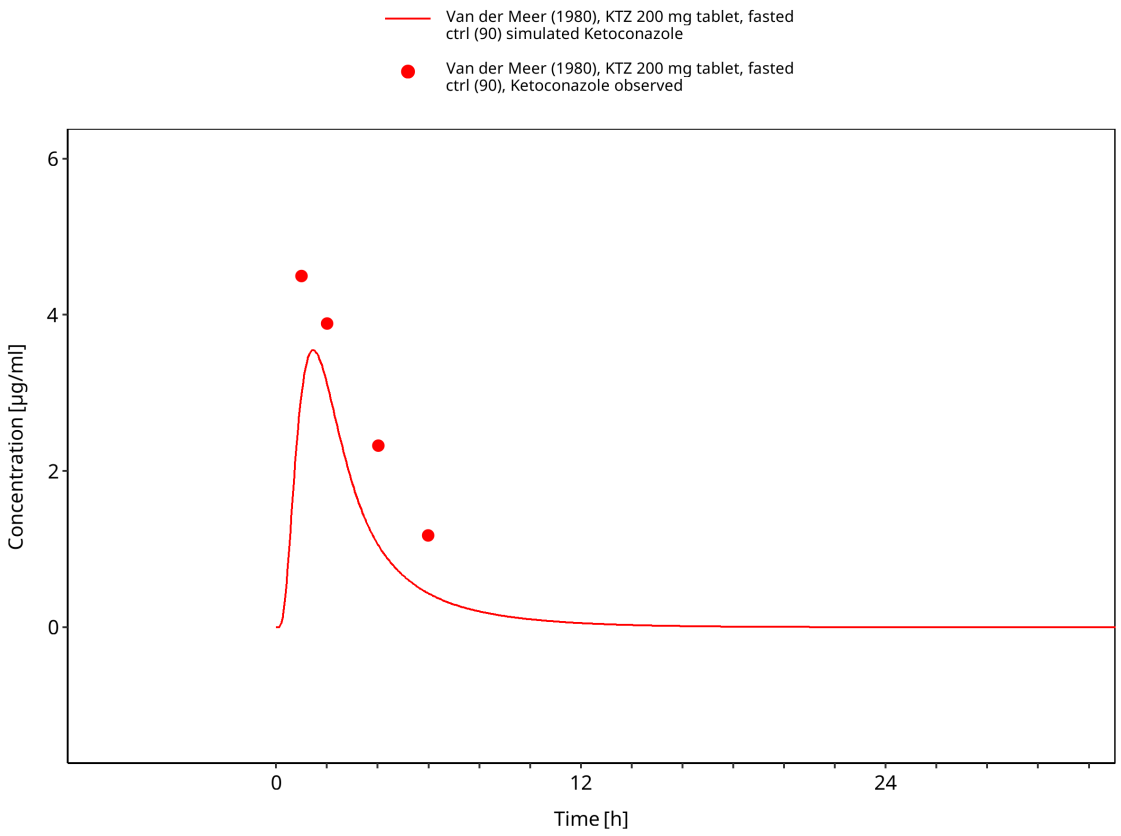


Figure 3-53: Van der Meer 1980: KTZ 200 mg tablet, fasted ctrl (90)

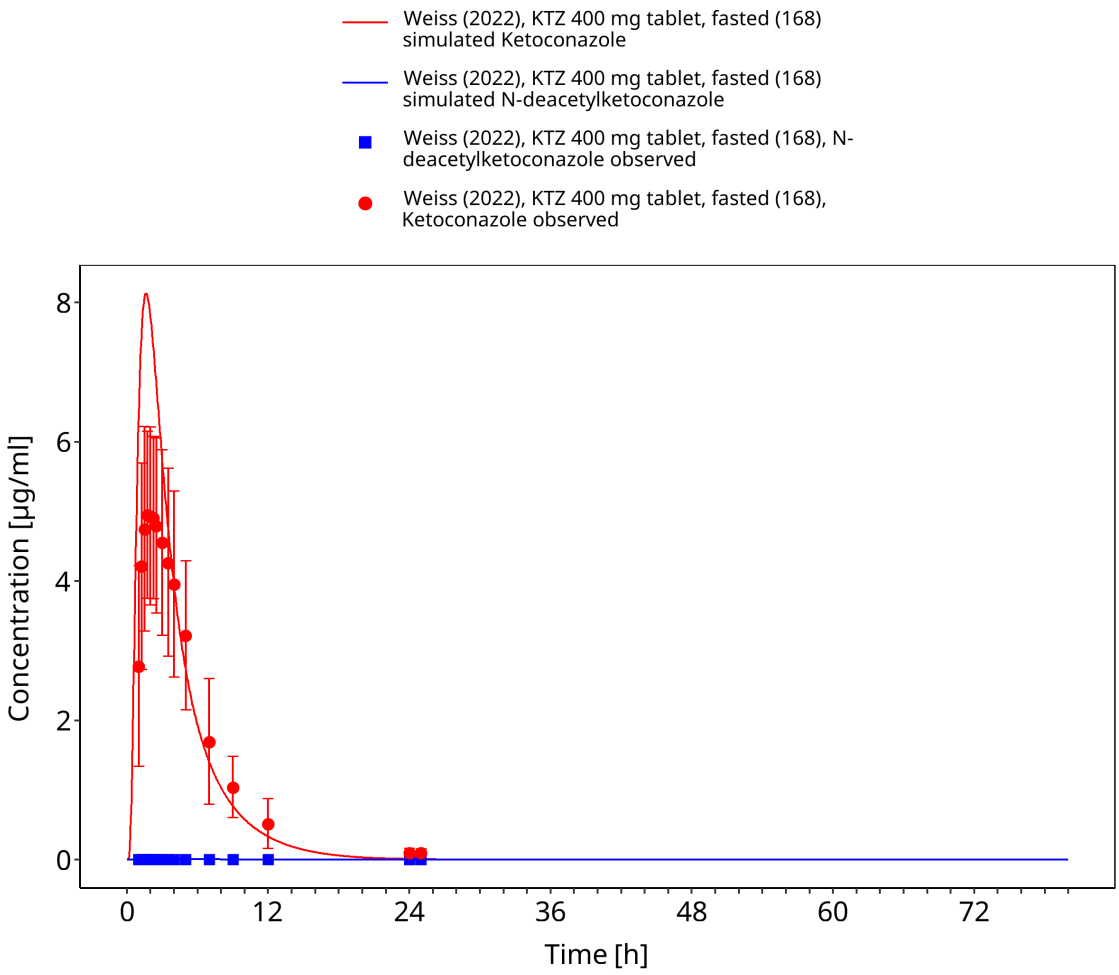


Figure 3-54: Weiss 2022: KTZ 400 mg tablet, fasted (168)

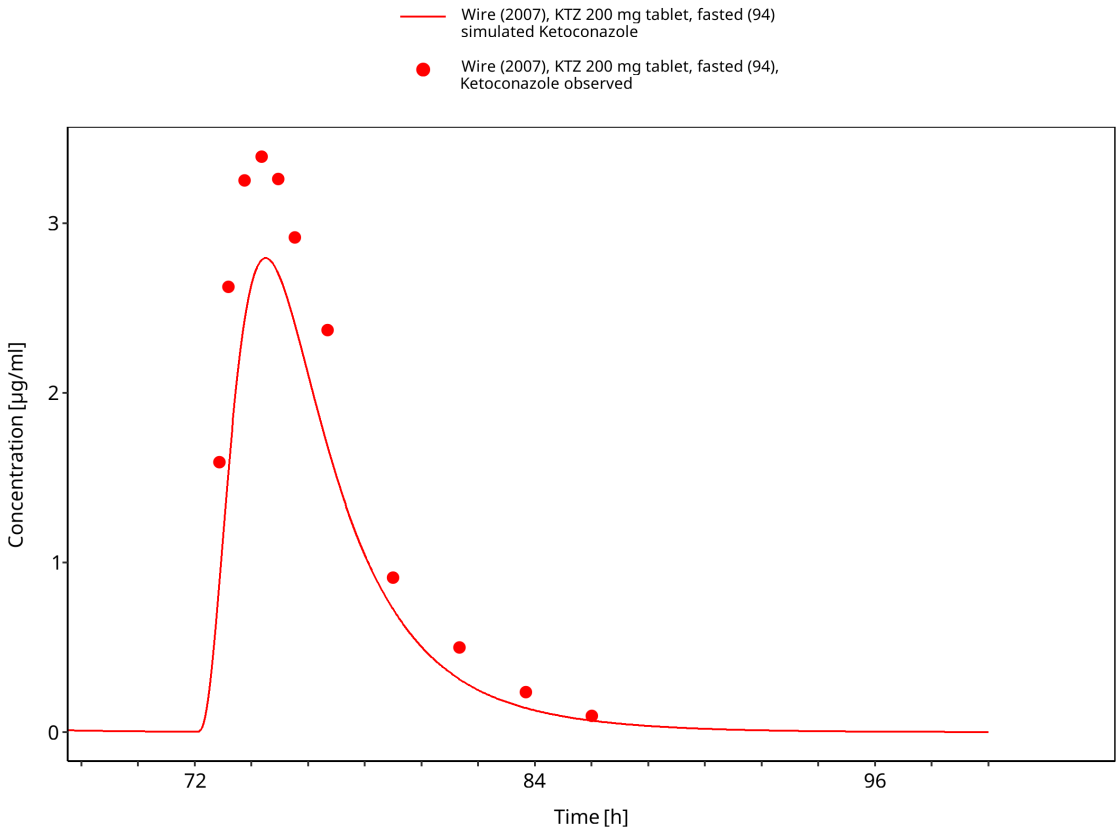


Figure 3-55: Wire 2007: KTZ 200 mg tablet q.d. 4d, fasted (94)

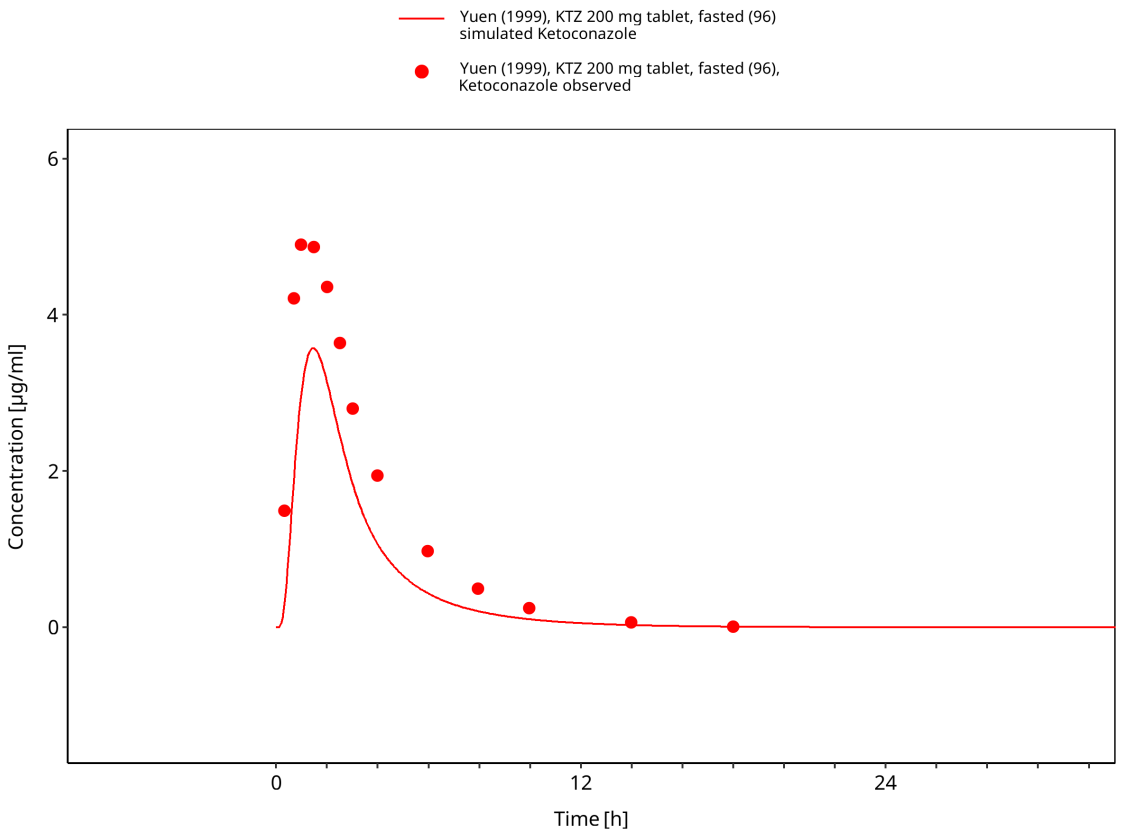


Figure 3-56: Yuen 1999: KTZ 200 mg tablet, fasted (96)

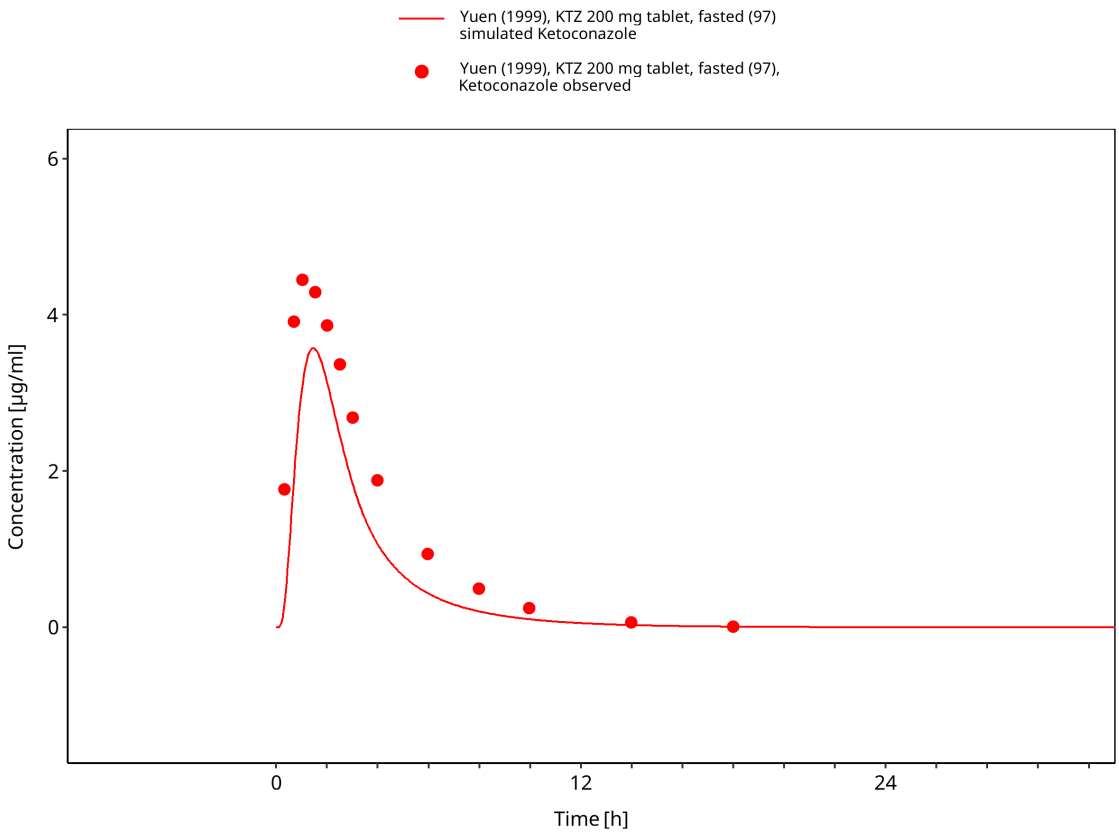


Figure 3-57: Yuen 1999: KTZ 200 mg tablet, fasted (97)

4 Conclusion

The ketoconazole parent-metabolite PBPK model describes the evaluated oral plasma concentration-time data across doses of 100-1200 mg in adults. The evaluation includes single-dose and multiple-dose regimens, solution and solid oral formulations, fasted and fed conditions, and ketoconazole and N-deacetylketoconazole observations.

The model combines formulation-dependent absorption with arylacetamide deacetylase, CYP3A4, UGT1A4, FMO3, and P-gp processes. Ketoconazole, N-deacetylketoconazole, and N-deacetyl-N-hydroxyketoconazole also contain competitive CYP3A4 and P-gp inhibition mechanisms.

The concentration-time profiles and goodness-of-fit diagnostics characterize model performance within the evaluated dose, regimen, formulation, food-condition, analyte, and adult population ranges. No formal acceptance criterion was applied. This compound report does not independently qualify DFI or DDI predictions.

The main limitations are the absence of intravenous data, estimated formulation and food-effect parameters, incomplete CYP3A4 and P-gp kinetic information, limited N-deacetylketoconazole data from one study, and no plasma data for N-deacetyl-N-hydroxyketoconazole.

5 References

[1] F Z Marok, J-G Wojtyniak, L M Fuhr, D Selzer, M Schwab, J Weiss, W E Haefeli, T Lehr. A Physiologically Based Pharmacokinetic Model of Ketoconazole and Its Metabolites as Drug-Drug Interaction Perpetrators. Pharmaceutics, 2023;15:679. doi: 10.3390/pharmaceutics15020679.

[2] J Weiss, K I Foerster, M Weber, J Burhenne, G Mikus, T Lehr, W E Haefeli. Does the circulating ketoconazole metabolite N-deacetyl ketoconazole contribute to the drug-drug interaction potential of the parent compound? Eur J Pharm Sci, 2022;169:106076. doi: 10.1016/j.ejps.2021.106076.

[3] T Fukami, A Iida, K Konishi, M Nakajima. Human arylacetamide deacetylase hydrolyzes ketoconazole to trigger hepatocellular toxicity. Biochem Pharmacol, 2016;116:153-161. doi: 10.1016/j.bcp.2016.07.007.

[4] W Fitch, T Tran, M Young, L Liu, Y Chen. Revisiting the Metabolism of Ketoconazole Using Accurate Mass. Drug Metab Lett, 2009;3:191-198. doi: 10.2174/187231209789352085.

[5] D Schwab, H Fischer, A Tabatabaei, S Poli, J Huwyler. Comparison of in Vitro P-Glycoprotein Screening Assays: Recommendations for Their Use in Drug Discovery. J Med Chem, 2003;46:1716-1725. doi: 10.1021/jm021012t.

[6] K Bourcier, R Hyland, S Kempshall, R Jones, J Maximilien, N Irvine, B Jones. Investigation into UDP-Glucuronosyltransferase (UGT) enzyme kinetics of imidazole- and triazole-containing antifungal drugs in human liver microsomes and recombinant UGT enzymes. Drug Metab Dispos, 2010;38:923-929. doi: 10.1124/dmd.109.030676.

[7] R C Heel, R N Brogden, A Carmine, P A Morley, T M Speight, G S Avery. Ketoconazole: A Review of its Therapeutic Efficacy in Superficial and Systemic Fungal Infections. Drugs, 1982;23:1-36. doi: 10.2165/00003495-198223010-00001.

[8] T K Daneshmend, D W Warnock, A Turner, C J C Roberts. Pharmacokinetics of ketoconazole in normal subjects. J Antimicrob Chemother, 1981;8:299-304. doi: 10.1093/jac/8.4.299.

[9] T K Daneshmend, D W Warnock. Clinical Pharmacokinetics of Ketoconazole. Clin Pharmacokinet, 1988;14:13-34. doi: 10.2165/00003088-198814010-00002.

[10] T K Daneshmend, D W Warnock, M D Ene, E M Johnson, G Parker, M D Richardson, C J C Roberts. Multiple dose pharmacokinetics of ketoconazole and their effects on antipyrine kinetics in man. J Antimicrob Chemother, 1983;12:185-188. doi: 10.1093/jac/12.2.185.

[11] T K Daneshmend, D W Warnock, M D Ene, E M Johnson, M R Potten, M D Richardson, P J Williamson. Influence of food on the pharmacokinetics of ketoconazole. Antimicrob Agents Chemother, 1984;25:1-3. doi: 10.1128/AAC.25.1.1.

[12] Y-C Huang, J L Colaizzi, R H Bierman, R Woestenborghs, J J P Heykants. Pharmacokinetics and dose proportionality of ketoconazole in normal volunteers. Antimicrob Agents Chemother, 1986;30:206-210. doi: 10.1128/AAC.30.2.206.

[13] R E Polk, M A Crouch, D S Israel, A Pastor, B M Sadler, G E Chittick, W T Symonds, W Gouldin, Y Lou. Pharmacokinetic interaction between ketoconazole and amprenavir after single doses in healthy men. Pharmacotherapy, 1999;19:1378-1384. doi: 10.1592/phco.19.18.1378.30905.

[14] M J Boyce, K J Baisley, S J Warrington. Pharmacokinetic interaction between domperidone and ketoconazole leads to QT prolongation in healthy volunteers: A randomized, placebo-controlled, double-blind, crossover study. Br J Clin Pharmacol, 2012;73:411-421. doi: 10.1111/j.1365-2125.2011.04093.x.

[15] P J Tiseo, C A Perdomo, L T Friedhoff. Concurrent administration of donepezil HCl and ketoconazole: Assessment of pharmacokinetic changes following single and multiple doses. Br J Clin Pharmacol, 1998;46 Suppl 1:30-34. doi: 10.1046/j.1365-2125.1998.0460s1030.x.

[16] P C Craven, J R Graybill, J H Jorgensen, W E Dismukes, B E Levine. High-dose ketoconazole for treatment of fungal infections of the central nervous system. Ann Intern Med, 1983;98:160-167. doi: 10.7326/0003-4819-98-2-160.

[17] P T Männistö, R Mäntylä, S Nykänen, U Lamminsivu, P Ottoila. Impairing effect of food on ketoconazole absorption. Antimicrob Agents Chemother, 1982;21:730-733. doi: 10.1128/AAC.21.5.730.

[18] D J Greenblatt, C E Wright, L L von Moltke, J S Harmatz, B L Ehrenberg, L M Harrel, K Corbett, M Counihan, S Tobias, R I Shader. Ketoconazole inhibition of triazolam and alprazolam clearance: Differential kinetic and dynamic consequences. Clin Pharmacol Ther, 1998;64:237-247. doi: 10.1016/S0009-9236(98)90172-2.

[19] T W F Chin, M Loeb, I W Fong. Effects of an acidic beverage (Coca-Cola) on absorption of ketoconazole. Antimicrob Agents Chemother, 1995;39:1671-1675. doi: 10.1128/AAC.39.8.1671.

[20] U.S. Food and Drug Administration. Bioequivalence, application number 74-971. 1998.

[21] C A Knupp, D C Brater, J Relue, R H Barbhaiya. Pharmacokinetics of didanosine and ketoconazole after coadministration to patients seropositive for the human immunodeficiency virus. J Clin Pharmacol, 1993;33:912-917. doi: 10.1002/j.1552-4604.1993.tb01921.x.

[22] H R Sadeghnia, M Hassanzadeh-Khayyat. Bioequivalency study of two formulations of ketoconazole tablet in healthy volunteers. Iran J Pharm Sci, 2005;1:209-215.

[23] W D S Solomon, P Senthamil Selvan, K Y G Gowda, U K Mandal, T K Pal. Evaluation of bioequivalence of two formulations containing 200 mg of ketoconazole. Asian J Chem, 2007;19:5365-5371.

[24] J W M Van der Meer, J J Keuning, H W Scheijgrond, J Heykants, J Van Cutsem, J Brugmans. The influence of gastric acidity on the bioavailability of ketoconazole. J Antimicrob Chemother, 1980;6:552-554. doi: 10.1093/jac/6.4.552.

[25] K H Yuen, J W Wong, N Billa, W P Choy, T Julianto. Comparative bioavailability study of two ketoconazole tablet preparations. Med J Malaysia, 1999;54:482-486.

[26] S C Piscitelli, T F Goss, J H Wilton, D T D'Andrea, H Goldstein, J J Schentag. Effects of ranitidine and sucralfate on ketoconazole bioavailability. Antimicrob Agents Chemother, 1991;35:1765-1771. doi: 10.1128/AAC.35.9.1765.

[27] S Sriwiriyajan, W Mahatthanatrakul, W Ridtitid, S Jaruratanasirikul. Effect of efavirenz on the pharmacokinetics of ketoconazole in HIV-infected patients. Eur J Clin Pharmacol, 2007;63:479-483. doi: 10.1007/s00228-007-0282-8.

[28] C G Patel, L Li, S Girgis, D M Kornhauser, E U Frevert, D W Boulton. Two-way pharmacokinetic interaction studies between saxagliptin and cytochrome P450 substrates or inhibitors: simvastatin, diltiazem extended-release, and ketoconazole. Clin Pharmacol Adv Appl, 2011;3:13-25. doi: 10.2147/CPAA.S15227.

[29] M B Wire, C H Ballow, J Borland, M J Shelton, Y Lou, G Yuen, J Lin, E W Lewis. Fosamprenavir plus ritonavir increases plasma ketoconazole and ritonavir exposure, while amprenavir exposure remains unchanged. Antimicrob Agents Chemother, 2007;51:2982-2984. doi: 10.1128/AAC.00008-07.

[30] V J Sekar, E Lefebvre, M De Pauw, T Vangeneugden, R M W Hoetelmans. Pharmacokinetics of darunavir/ritonavir and ketoconazole following co-administration in HIV-negative healthy volunteers. Br J Clin Pharmacol, 2008;66:215-221. doi: 10.1111/j.1365-2125.2008.03191.x.

[31] Chemicalize. Ketoconazole entry. 2021. Accessed 2021-05-01.

[32] H S Ghazal, A M Dyas, J L Ford, G A Hutcheon. The impact of food components on the intrinsic dissolution rate of ketoconazole. Drug Dev Ind Pharm, 2015;41:1647-1654. doi: 10.3109/03639045.2014.983114.

[33] ChemSpider. Ketoconazole entry. Accessed 2021-05-01. https://www.chemspider.com/Chemical-Structure.401695.html.

[34] Open Systems Pharmacology Suite Community. Open Systems Pharmacology Suite Manual. 2018. https://docs.open-systems-pharmacology.org/.

[35] R S Fisher, E Rock, L S Malmud. Effects of meal composition on gallbladder and gastric emptying in man. Dig Dis Sci, 1987;32:1337-1344. doi: 10.1007/BF01296658.

[36] L M Berezhkovskiy. Volume of distribution at steady state for a linear pharmacokinetic system with peripheral elimination. J Pharm Sci, 2004;93:1628-1640. doi: 10.1002/jps.20073.

[37] E J Elder, J C Evans, B D Scherzer, J E Hitt, G B Kupperblatt, S A Saghir, D A Markham. Preparation, characterization, and scale-up of ketoconazole with enhanced dissolution and bioavailability. Drug Dev Ind Pharm, 2007;33:755-765. doi: 10.1080/03639040601031882.

[38] Chemicalize. N-deacetylketoconazole entry. 2021b. Accessed 2021-05-01.

[39] R Kawai, M Lemaire, J L Steimer, A Bruelisauer, W Niederberger, M Rowland. Physiologically based pharmacokinetic study on a cyclosporin derivative, SDZ IMM 125. J Pharmacokinet Biopharm, 1994;22:327-365. doi: 10.1007/BF02353860.

[40] T Rodgers, M Rowland. Physiologically based pharmacokinetic modelling 2: predicting the tissue distribution of acids, very weak bases, neutrals and zwitterions. J Pharm Sci, 2006;95:1238-1257. doi: 10.1002/jps.20502.

[41] R J Rodriguez, D Acosta. Metabolism of ketoconazole and deacetylated ketoconazole by rat hepatic microsomes and flavin-containing monooxygenases. Drug Metab Dispos, 1997;25:772-777.

[42] Chemicalize. N-deacetyl-N-hydroxyketoconazole entry. 2022. Accessed 2021-05-01.

1. The model stores the ketoconazole solubility relationship as a table function. The scalar displayed in the generated parameter table is an internal representation of that function and should not be interpreted as the solubility at a specific pH.↩